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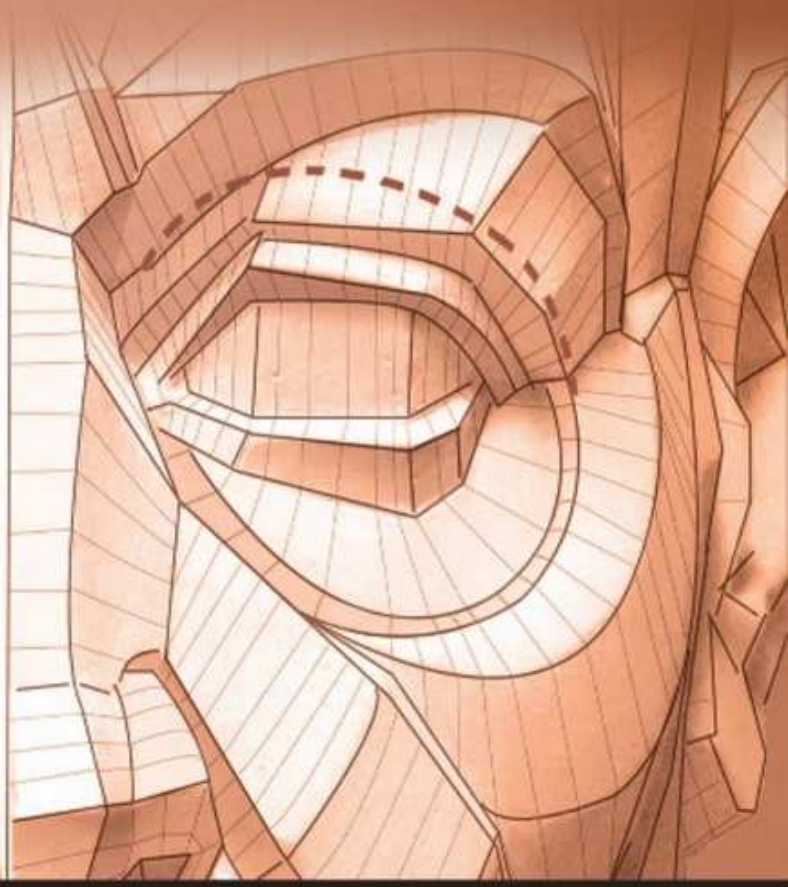
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Anatomy Q-Bank

Head and Neck

Hosam Eldeen Elsadig Gasmalla





First Edition 2016

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Dedication

To the soul of my grandfather: Gasmalla, and my Aunt: Majida

Preface

This book is part of a series that is designed to help medical student to study via solving questions, to get the maximum benefit, one shall study from a rich and knowledgeable textbook first; the questions in this book are grouped into four sections

- **Section One: MCQs (Type-A): covers the level of recall in the cognitive domain.**
- **Section Two: Scenario-Based Questions: the questions are primary MCQs based on given scenarios (Cases) to cover the level of comprehension in the cognitive domain.**
- **Section Three: Extended Matching Questions EMQs to cover the level of the application of knowledge in the cognitive domain.**
- **Section Four: Medical Imaging: questions about normal X-ray and CT scan images are provided in this section.**

I'm looking forward to the useful and productive feedback from my fellow colleagues and the students.

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Acknowledgement

I would like to express my gratitude and appreciation to the generations' teacher: Dr. Abbas Gareeballa who has been helping, supporting and advising me since I was a medical student.

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Questions

Section One (MCQs A- Type)

Skull and Cervical Vertebrae

1. In the anterior view of the viscerocranium, which one of the following bones is single?

- **a.** Nasal.
- **b.** Frontal.
- **c.** Lacrimal.
- **d.** Maxillary.
- **e.** Zygomatic.

2. What is the strongest bone among the bones of the face?

- **a.** Nasal.
- **b.** Frontal.
- **c.** Maxillary.
- **d.** Zygomatic.
- **e.** Mandibular.

3. The superior orbital margin is formed by the ____ bone:

- **a.** Frontal.
- **b.** Maxillary.
- **c.** Zygomatic.
- **d.** Ethmoidal.
- **e.** Sphenoidal.

4. Which part of the frontal bone contributes to the floor of the anterior cranial fossa?

- **a.** Nasal.
- **b.** Squamous.
- **c.** Orbital plate.
- **d.** Maxillary process.
- **e.** Zygomatic process.

5. Which part of the frontal bone articulates at the Pterion?

- **a.** Nasal.
- **b.** Squamous.
- **c.** Orbital plate.
- **d.** Maxillary process.
- **e.** Zygomatic process.

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6. The metopic suture of the frontal bone normally disappears at about:

- **a.** 2 months.
- **b.** 4 months.
- **c.** 5 years.
- **d.** 8 years.
- **e.** 15 years.

7. In the anterior cranial fossa, the cribriform plate is part of the ___ bone:

- **a.** Frontal.
- **b.** Maxillary.
- **c.** Zygomatic.
- **d.** Ethmoidal.
- **e.** Sphenoidal.

8. In the walls of the orbit, the ethmoid bone contributes to the formation of the:

- **a.** Medial orbital wall.
- **b.** Inferior orbital wall.
- **c.** Lateral orbital margin.
- **d.** Superior orbital margin.
- **e.** Floor of the orbital cavity.

9. The ___ artery exits the middle cranial fossa through the optic canal:

- **a.** Buccal.
- **b.** Pterygoid.
- **c.** Masseteric.
- **d.** Ophthalmic.
- **e.** Deep temporal.

10. The Posterolateral fontanelle is located at the:

- **a.** Inion.
- **b.** Pterion.
- **c.** Bregma.
- **d.** Lambda.
- **e.** Asterion.

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11. The presence of the fontanelles is due to the delayed ossification of the four angels of the ___ bone:

- **a.** Frontal.
- **b.** Parietal.
- **c.** Occipital.
- **d.** Sphenoid.
- **e.** Temporal.

12. The nerve that provides motor supply to the tongue emerges from the cranium through a canal in the ___ bone:

- **a.** Nasal.
- **b.** Maxilla.
- **c.** Parietal.
- **d.** Lacrimal.
- **e.** Occipital.

13. What is the foramen located between the occipital bone and petrous part of temporal bone?

- **a.** Condylar canal.
- **b.** Jugular foramen.
- **c.** Mastoid foramen.
- **d.** Foramen magnum.
- **e.** Hypoglossal canal.

14. The internal acoustic meatus is a canal within the ____ bone:

- **a.** Occipital.
- **b.** Maxillary.
- **c.** Temporal.
- **d.** Sphenoid.
- **e.** Zygomatic.

15. The acoustic labyrinth is situated within the ___ part of temporal bone:

- **a.** Petrous.
- **b.** Squamous.
- **c.** Tympanic.
- **d.** Styloid process.
- **e.** Zygomatic process.

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16. The superior orbital fissure is a gap within the ___ bone:

- **a.** Frontal.
- **b.** Parietal.
- **c.** Occipital.
- **d.** Sphenoid.
- **e.** Temporal.

17. What is the foramen located between the sphenoid bone and petrous part of temporal bone?

- **a.** Foramen ovale.
- **b.** Jugular foramen.
- **c.** Foramen lacerum.
- **d.** Foramen rotundum.
- **e.** Foramen spinosum.

18. The internal carotid artery passes across the foramen _____ to the cranium:

- **a.** Ovale.
- **b.** Lacerum.
- **c.** Magnum.
- **d.** Spinosum.
- **e.** Rotundum.

19. Foramen spinosum is located in the __ of sphenoidal bone:

- **a.** Body.
- **b.** Lesser wing.
- **c.** Sella turcica.
- **d.** Greater wing.
- **e.** Pterygoid plates.

20. What is the nerve that passes through foramen rotundum?

- **a.** Olfactory.
- **b.** Maxillary.
- **c.** Mandibular.
- **d.** Ophthalmic.
- **e.** Vestibulocochlear.

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21. The mandibular division of the trigeminal nerve exits the skull through:

- **a.** Foramen ovale.
- **b.** Foramen lacerum.
- **c.** Foramen spinosum.
- **d.** Foramen rotundum.
- **e.** Superior orbital fissure.

22. Which part of sphenoid bone ossifies by intramembranous ossification?

- **a.** Body.
- **b.** Lesser wing.
- **c.** Sella turcica.
- **d.** Greater wing.
- **e.** Pterygoid plates.

23. Scaphocephaly results from premature closure of the ___ suture:

- **a.** Sagittal.
- **b.** Coronal.
- **c.** Lambdoid.
- **d.** Parietomastoid.
- **e.** Sphenosquamosal.

24. The Axis is an atypical cervical vertebra because of the:

- **a.** Lack of body.
- **b.** Presence of longest spine.
- **c.** Presence of the odontoid process.
- **d.** Absence of foramen transversarium.
- **e.** Presence of lateral mass on each side.

25. The absence of a body makes the ___ vertebra atypical:

- **a.** 1st cervical.
- **b.** 2nd cervical.
- **c.** 3rd thoracic.
- **d.** 4th thoracic.
- **e.** 5th lumbar.

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26. The foramen transversarium of the _____ cervical vertebra does not transmit the vertebral artery.

a. 1st.
b. 2nd.
c. 4th.
d. 6th.
e. 7th.

27. What is the axis of the Atlantoaxial Joints during their movement?

a. Odontoid process.
b. Occipital condyles.
c. Lateral mass of C1.
d. Lateral mass of C2.
e. Anterior arch of C1.

28. The hyoid bone is connected to the scapula through ___ muscle:

a. Omohyoid.
b. Mylohyoid.
c. Stylohyoid.
d. Geniohyoid.
e. Sternohyoid.

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Neck

29. Which one of the following vessels is located just deep to platysma?

- **a.** External jugular vein.
- **b.** External carotid artery.
- **c.** Common carotid artery.
- **d.** Internal carotid artery.
- **e.** Internal jugular vein.

30. The platysma is innervated by the ____ nerve:

- **a.** Facial
- **b.** Accessory
- **c.** Trigeminal
- **d.** Glossopharyngeal
- **e.** Transverse cervical

31. Which one of the following ligaments is a thickened part of investing layer of the deep cervical fascia?

- **a.** Thyrohyoid.
- **b.** Spuraspinous.
- **c.** Cricothyroid.
- **d.** Stylomandibular.
- **e.** Ligamentum flaveum.

32. Sternocleidomastoid muscle is completely surrounded by the:

- **a.** Carotid sheath.
- **b.** Investing layer.
- **c.** Superficial fascia.
- **d.** Pretracheal fascia.
- **e.** Prevertebral fascia.

33. The ligament that suspends the hyoid bone to digastric muscle is derived from _____?

- **a.** Carotid sheath.
- **b.** Superficial fascia.
- **c.** Pretracheal fascia.
- **d.** Prevertebral fascia.
- **e.** Thyrohyoid membrane.

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34. Which one of the following ligaments of the neck limits hyperextension?

- **a.** Nuchal ligament.
- **b.** Ligamentum flavum.
- **c.** Interspinous ligaments.
- **d.** Intertransverse ligaments.
- **e.** Anterior longitudinal ligament.

35. What is the layer that overlies the phrenic nerve?

- **a.** Platysma.
- **b.** Superficial fascia.
- **c.** Pretracheal fascia.
- **d.** Prevertebral fascia.
- **e.** Investing layer of deep fascia.

36. Pus in _____ pace of the neck passes as downward as the superior mediastinum.

- **a.** Submental.
- **b.** Prevertebral.
- **c.** Submandibular.
- **d.** Parapharyngeal.
- **e.** Reteropharyngeal.

37. Moving the head so that the face looks upwards and to the left is a movement produced by the:

- **a.** Platysma.
- **b.** Left trapezius only.
- **c.** Right geniohyoid only.
- **d.** Right sternocleidomastoid.
- **e.** Posterior belly of digastric.

38. The neck is divided into anterior and posterior triangles using ____ muscle as land mark:

- **a.** Trapezius.
- **b.** Sternohyoid.
- **c.** Scalenus anterior
- **d.** Sternocleidomastoid.
- **e.** Anterior belly of digastric.

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39. The _____ triangle of the neck lies between the anterior bellies of digastric muscles.

a. Occipital.
b. Posterior.
c. Muscular.
d. Submental.
e. Submandibular.

40. The posterior triangle is divided into two triangles by the ___ muscle:

a. Digastric.
b. Trapezius.
c. Omohyoid.
d. Mylohyoid.
e. Sternohyoid.

41. Stab wound directed into submandibular triangle may lead to:

a. Ptosis of the eye.
b. Drop of the shoulder.
c. Deviation of the mouth.
d. Deviation of the tongue.
e. Tilting of the head to the same side.

42. Roots of cervical plexus pass anterior to _____ muscle.

a. Scalenus medius.
b. Scalenus anterior.
c. Scalenus posterior.
d. Semispinalis capitis.
e. Sternocleidomastoid.

43. The nerve point of the neck is at the middle of the posterior border of the _____ muscle.

a. Trapezius.
b. Omohyoid.
c. Sternohyoid.
d. Sternothyroid.
e. Sternocleidomastoid.

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44. What is the cutaneous branch of the cervical plexus that crosses over the sternocleidomastoid muscle?

- **a.** Phrenic.
- **b.** Great auricular.
- **c.** Supraclavicular.
- **d.** Lesser occipital.
- **e.** Transverse cervical.

45. The branch from the hypoglossal nerve that supplies thyrohyoid muscle is actually derived from _____

- **a.** C1.
- **b.** C2.
- **c.** C3.
- **d.** C4.
- **e.** C5.

46. Inferior root of Ansa hypoglossi is derived from:

- **a.** C1.
- **b.** C2 and C3.
- **c.** C4 and C5.
- **d.** C6 and C7.
- **e.** C8.

47. The inferior root of ansa cervicalis spirals around the:

- **a.** Subclavian artery.
- **b.** Internal jugular vein.
- **c.** External jugular vein.
- **d.** Internal carotid artery.
- **e.** External carotid artery.

48. Which one of the following branches from the cervical plexus is mixed (motor and sensory)?

- **a.** Phrenic.
- **b.** Great auricular.
- **c.** Supraclavicular.
- **d.** Lesser occipital.
- **e.** Transverse cervical.

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49. The _____ nerve from the cervical plexus supplies the skin over the acromion process of the scapula:

a. Phrenic.
b. Supraclavicular.
c. Great auricular.
d. Lesser occipital.
e. Transverse cervical.

50. A patient had lost cutaneous sensation over the angle of the mandible, which nerve is most likely damaged?

a. Mandibular.
b. Ophthalmic.
c. Great auricular.
d. Auricotemporal.
e. Supra clavicular.

51. The thyroid gland moves with swallowing because it is tight by the ___ layer of deep cervical fascia:

a. Parotid.
b. Carotid.
c. Investing.
d. Pretracheal.
e. Prevertebral.

52. The superior thyroid artery is closely related to the ____ nerve:

a. Lingual.
b. Hypoglossal.
c. Internal laryngeal.
d. Glossopharyngeal.
e. Recurrent laryngeal.

53. Miosis in Horner's syndrome is due to lack of innervation from the _____ ganglion:

a. Otic.
b. Ciliary.
c. Cervical.
d. Submandibular.
e. Pterygopalatine.

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54. Ptosis in Horner's syndrome is due to paralysis of the smooth muscle part of the _____ muscle:

a. Superior rectus.
b. Superior oblique.
c. Depressor anguli oris.
d. Levator palpebrae superioris.
e. Levator labii superioris alaeque nasii.

55. Damage to _____ ganglion will cause xerophthalmia:

a. Otic.
b. Ciliary.
c. Stellate.
d. Submandibular.
e. Pterygopalatine.

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Face and SCALP

56. Cutaneous innervation of the forehead is provided by the ____ nerve:

- **a.** Facial.
- **b.** Infraorbital.
- **c.** External nasal.
- **d.** Supratrochlear.
- **e.** Zygomaticofacial.

57. Loss of sensation of skin over the tip of the nose is due to injury of _____ nerve.

- **a.** Facial.
- **b.** Olfactory.
- **c.** Maxillary.
- **d.** Mandibular.
- **e.** Ophthalmic.

58. The cutaneous innervation of the anterior part of the SCALP is derived from branches of the _____ nerve:

- **a.** Facial.
- **b.** Maxillary.
- **c.** Ophthalmic.
- **d.** Great auricular.
- **e.** Lesser occipital.

59. Among the nerves of the face, which one of the following is a direct continuation of the maxillary nerve?

- **a.** Lacrimal
- **b.** Infraorbital.
- **c.** Supraorbital.
- **d.** Supratrochlear.
- **e.** Infratrochlear.

60. The ____ muscle compresses cheek against molar teeth:

- **a.** Risorius.
- **b.** Mentalis.
- **c.** Buccinator.
- **d.** Orbicularis oris.
- **e.** Zygomaticus major.

61. The _____ muscle draws a smile on the face:

- **a.** Risorius.
- **b.** Levator anguli oris.
- **c.** Zygomaticus minor.
- **d.** Corrugator supercilii.
- **e.** Levator labii superioris.

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62. The _____ muscle helps to produce the expression of surprise on the face:

- **a.** Platysma.
- **b.** Buccinator.
- **c.** Occipitofrontalis.
- **d.** Orbicularis oculi.
- **e.** Corrugator supercilii.

63. Parasympathetic innervation to the parotid gland is provided through the _____ ganglion:

- **a.** Otic.
- **b.** Ciliary.
- **c.** Geniculate.
- **d.** Submandibular.
- **e.** Pterygopalatine.

64. Sensory innervation of the parotid sheath is carried by ___ nerve:

- **a.** Facial.
- **b.** Vidian.
- **c.** Maxillary.
- **d.** Great auricular.
- **e.** Glossopharyngeal.

65. Sensory innervation of the parotid gland is carried by _____ nerve:

- **a.** Vagus.
- **b.** Buccal.
- **c.** Great auricular.
- **d.** Auriculotemporal.
- **e.** Glossopharyngeal.

66. The preganglionic fibers to the otic ganglion reach it through the _____ nerve:

- **a.** Vidian.
- **b.** Deep petrosal.
- **c.** Lesser petrosal.
- **d.** Greater petrosal.
- **e.** Lesser occipital.

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Temporal, infratemporal and pterygopalatine fossae

67. The _____ muscle retracts the mandible:

- **a.** Masseter.
- **b.** Temporalis.
- **c.** Buccinator.
- **d.** Medial pterygoid
- **e.** Lateral pterygoid.

68. Medial pterygoid muscle is inserted into the medial surface of the:

- **a.** Zygomatic bone.
- **b.** Medial pterygoid plate.
- **c.** Lateral pterygoid plate.
- **d.** Ramus of the mandible.
- **e.** Coronoid process of the mandible.

69. Which one of the following muscles of mastication opens the mouth?

- **a.** Masseter.
- **b.** Temporalis.
- **c.** Buccinator.
- **d.** Medial pterygoid
- **e.** Lateral pterygoid.

70. Masseter muscle is inserted into the _____ of the mandible:

- **a.** Neck.
- **b.** Angle.
- **c.** Mylohyoid line.
- **d.** Coronoid process.
- **e.** Medial surface of ramus.

71. Which one of the following branches of the mandibular nerve supplies the anterior belly of Digastric muscle:

- **a.** Buccal.
- **b.** Masseteric.
- **c.** Deep temporal.
- **d.** Nerve to Mylohyoid.
- **e.** Nerve to Lateral pterygoid.

72. The sensory supply of the chin is derived from the _____ nerve:

- **a.** Buccal.
- **b.** Lingual.
- **c.** Meningeal.
- **d.** Inferior alveolar.
- **e.** Auriculotemporal.

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73. The pterygopalatine fossa is bounded superiorly by the:

- **a.** Zygomatic arch.
- **b.** Greater wing of the sphenoid.
- **c.** Posterior aspect of the maxilla.
- **d.** Pterygoid process of the sphenoid.
- **e.** Perpendicular plate of the palatine bone.

74. The maxillary nerve enters the pterygopalatine fossa through ____:

- **a.** Foramen ovale.
- **b.** Foramen rotundum.
- **c.** Inferior orbital fissure.
- **d.** Pterygomaxillary fissure.
- **e.** Sphenopalatine foramen.

75. The maxillary nerve is connected to the Ophthalmic nerve through the ___ nerve:

- **a.** Ganglionic.
- **b.** Infraorbital.
- **c.** Inferior Alveolar.
- **d.** Zygomatico-temporal.
- **e.** Posterior Superior Alveolar.

76. The preganglionic parasympathetic root of the Pterygopalatine ganglion is the _____ nerve:

- **a.** Deep petrosal.
- **b.** Lesser occipital.
- **c.** Greater petrosal.
- **d.** Greater occipital.
- **e.** Superficial petrosal.

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Oral Cavity and palate

77. Which one of the following muscles forms the diaphragm of the oral cavity?

a. Digastric.
b. Mylohyoid.
c. Genohyoid.
d. Stylohyoid.
e. Genoglossus.

78. Which one of the following structures is located inferior to mylohyoid muscle?

a. Facial artery.
b. Lingual nerve.
c. Sublingual gland.
d. Geniohyoid muscle.
e. Submandibular duct.

79. Regarding hypoglossal nerve injury (at one side), the deviation of the tongue is produced mainly due to the paralysis of _____ muscle:

a. Stylohyoid.
b. Hyoglossus.
c. Mylohyoid.
d. Genioglossus.
e. Palatoglossus.

80. Which one of the following muscles of the tongue narrows it?

a. Hyoglossus.
b. Genioglossus.
c. Palatoglossus.
d. Intrinsic vertical.
e. Intrinsic Transverse.

81. Special sensation from the tip of the tongue is carried by the _____ nerve:

a. Vagus.
b. Lingual.
c. Hypoglossal.
d. Chorda tympani.
e. Glossopharyngeal.

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82. The ____ muscle of the tongue is innervated by the pharyngeal plexus:

- **a.** Hyoglossus.
- **b.** Styloglossus.
- **c.** Palatoglossus.
- **d.** Intrinsic vertical.
- **e.** Intrinsic Transverse.

83. Which one of the following muscles raises the floor of the mouth in the first stage of swallowing?

- **a.** Digastric.
- **b.** Mylohyoid.
- **c.** Palatopharyngeus.
- **d.** Tensor veli palatine.
- **e.** Levator veli palatine.

84. In the voluntary stage of swallowing, the bolus is compressed against the palate mainly by the action of _____ muscle:

- **a.** Palatopharyngeus
- **b.** Middle constrictor.
- **c.** Tensor veli palatine.
- **d.** Superior constrictor.
- **e.** Levator veli palatine.

85. Which one of the following muscles of the palate is innervated by the trigeminal nerve?

- **a.** Palatoglossus.
- **b.** Musculus uvulae.
- **c.** Palatopharyngeus.
- **d.** Tensor veli palatine.
- **e.** Levator veli palatine.

86. Nasopalatine nerve block affects the alveolar bone of the:

- **a.** Upper molars.
- **b.** Upper incisors.
- **c.** Lower incisors.
- **d.** Lower canines.
- **e.** Lower premolars.

87. The palatine tonsil is normally found within the_____.

- **a.** Valeculla.
- **b.** Oro-pharynx.
- **c.** Piriform fossa.
- **d.** Naso-pharynx.
- **e.** Laryngo-pharynx.

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88. The lingual tonsils are normally found within the_____.

a. Valeculla.
b. Oro-pharynx.
c. Piriform fossa.
d. Naso-pharynx.
e. Laryngo-pharynx.

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Nasal Cavity and Paranasal Sinuses

89. What is the bone that forms major parts of roof, lateral walls and septum of the nasal cavity, all at the same time?

a. Vomer.
b. Maxilla.
c. Ethmoid.
d. Sphenoid.
e. Palatine bone.

90. The superior concha is part of_____ bones.

a. Frontal.
b. Palatine.
c. Sphenoid.
d. Maxillary.
e. Ethmoidal.

91. The bone that forms parts of both lateral wall and floor of the nasal cavity is:

a. Vomer.
b. Frontal.
c. Maxilla.
d. Sphenoid.
e. Inferior concha.

92. The nasal and cranial cavities are communicated together through:

a. Pterygoid canal.
b. Cribriform plate.
c. Incisive foramen.
d. Lesser palatine foramen.
e. Pterygo-maxillary fissure.

93. Which one of the following opens into the inferior meatus of the nasal cavity?

a. Maxillary sinus.
b. Sphenoidal sinus.
c. Naso-lacrimal Duct.
d. Posterior ethmoidal.
e. Middle ethmoidal cells.

94. The ____ artery is main artery of the nasal cavity:

a. Inferior labial.
b. Superior labial.
c. Lesser palatine.
d. Sphenopalatine.
e. Greater palatine.

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95. The common site of nosebleed, (Little's area) is located at the_____of the nose.

- **a.** Roof.
- **b.** Floor.
- **c.** Septum.
- **d.** Left lateral wall.
- **e.** Right lateral wall.

96. General sensation from the roof the nasal cavity is carried by_____nerve.

- **a.** Facial
- **b.** Olfactory.
- **c.** Maxillary.
- **d.** Mandibular.
- **e.** Ophthalmic.

97. The special sensation from the upper part of the nasal septum is carried by_____ nerve.

- **a.** Facial
- **b.** Olfactory.
- **c.** Maxillary.
- **d.** Mandibular.
- **e.** Ophthalmic.

98. The mucous membrane of the inferior concha is innervated by_____ nerve.

- **a.** Facial.
- **b.** Olfactory.
- **c.** Maxillary.
- **d.** Mandibular.
- **e.** Ophthalmic.

99. The sphenoidal sinus is related laterally to the:

- **a.** Pons.
- **b.** Pituitary fossa.
- **c.** Medulla oblongata.
- **d.** Internal carotid artery.
- **e.** Roof of the Nasopharynx.

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100. Sensory innervation of the sphenoidal sinus is provided by the _____ nerve:

- **a.** Nasociliary.
- **b.** Supraorbital.
- **c.** Posterior ethmoidal.
- **d.** Anterior superior alveolar.
- **e.** Posterior superior alveolar.

101. The Base of the maxillary sinus forms the:

- **a.** Hard palate.
- **b.** Floor of the orbit.
- **c.** Floor of the nasal cavity.
- **d.** Alveolar part of the maxilla.
- **e.** Lateral wall of the nasal cavity.

102. The adenoid is located within the _____.

- **a.** Valeculla.
- **b.** Oro-pharynx.
- **c.** Naso-pharynx.
- **d.** Piriform fossa.
- **e.** Laryngo-pharynx.

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Pharynx and Larynx

103. The pharynx extends lower down to the level of ____ vertebra:

- **a.** 3rd cervical.
- **b.** 4th cervical.
- **c.** 6th cervical.
- **d.** 1st thoracic.
- **e.** 4th thoracic.

104. The hypopharynx is located posterior to the:

- **a.** Larynx.
- **b.** Soft palate.
- **c.** Oral cavity.
- **d.** Nasal cavity.
- **e.** Nasal septum.

105. The roof of the nasopharynx is built up by two bones: the basilar part of the occipital bone and the _____ bone:

- **a.** Palatine.
- **b.** Lacrimal.
- **c.** Maxillary.
- **d.** Ethmoidal.
- **e.** Sphenoidal.

106. The pharyngeal muscle that originates from the pterygomandibular raphe is the:

- **a.** Stylo-pharyngeus.
- **b.** Middle constrictor.
- **c.** Palato-pharyngeus.
- **d.** Superior constrictor.
- **e.** Salpingo-pharyngeus.

107. The pharyngeal muscle that is innervated solely by the glossopharyngeal nerve is the _____ muscle:

- **a.** Stylopharyngeus.
- **b.** Palatopharyngeus
- **c.** Middle constrictor.
- **d.** Salpingopharyngeus.
- **e.** Superior constrictor.

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108. The upper border of thyroid cartilage lies at level with _____ vertebra:

a. C 2.
b. C 4.
c. C6.
a. T2.
b. T4.

109. The cricoid cartilage lies at level with …. vertebra:
a. C 4.
b. C6.
c. T2.
d. T4.
e. T6.

110. The type of joint in cricothyroid joint is:
a. Fibrous.
b. Typical synovial.
c. Atypical synovial.
d. Primary cartilaginous.
e. Secondary cartilaginous.

111. The laryngeal vestibule is located:
a. Below the vocal folds.
b. Above the vestibular folds.
c. Above the aryepiglottic fold.
d. Between the vestibular and vocal folds.
e. Lateral to the middle part of the laryngeal

112. Which one of the following muscles elevates the hyoid bone during swallowing?
a. Omohyoid.
b. Genohyoid.
c. Thyrohyoid.
d. Sternohyoid.
e. Sternothyroid.

113. Which one of the following muscles abducts the vocal folds?
a. Cricothyroid.
b. Thyroarytenoid.
c. Transverse arytenoids.
d. Lateral cricoarytenoid.
e. Posterior cricoarytenoid.

114. The cricothyroid muscle is innervated by the __ nerve:
a. Hypoglossal.

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b. Internal laryngeal.
c. Glossopharyngeal.
d. External laryngeal.
e. Recurrent laryngeal.

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Ear

115. Among the depressions of the auricle of the ear, the ------ is the deepest one:

- **a.** Scapha.
- **b.** Concha.
- **c.** Scaphoid fossa.
- **d.** Triangular fossa.
- **e.** Intertragic notch.

116. The sensory innervation of the medial surface of the auricle (back of the ear) is provided by the _____ nerve:

- **a.** Facial.
- **b.** Ophthalmic.
- **c.** Great auricular.
- **d.** Greater occipital.
- **e.** Auriculotemporal.

117. "Pars interna" of the external acoustic meatus is directed:

- **a.** Anterolaterally.
- **b.** Posterolaterally.
- **c.** Anteromedially.
- **d.** Posteromedially.
- **e.** Superiomedially.

118. The tympanic membrane is at the ….. wall of the middle ear:

- **a.** Medial.
- **b.** Lateral.
- **c.** Inferior.
- **d.** Anterior.
- **e.** Posterior.

119. The roof of the middle ear is formed by the ….. component of the temporal bone:

- **a.** Petrous part.
- **b.** Squamous part.
- **c.** Tympanic part.
- **d.** Styloid process.
- **a.** Zygomatic process.

120. The pharyngotympanic tube is at the ….. wall of the middle ear:

- **a.** Medial.
- **b.** Lateral.
- **c.** Inferior.
- **d.** Anterior.
- **e.** Posterior.

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121. The _____ is closely related to the floor of the middle ear:

- **a.** Internal jugular vein.
- **b.** Anterior jugular vein.
- **c.** Retromandibular vein.
- **d.** Internal carotid artery.
- **e.** Common carotid artery.

122. The tensor tympani muscle is attached to the:

- **a.** Incus.
- **b.** Stapes.
- **c.** Malleus.
- **d.** Round window.
- **e.** Tympanic membrane.

123. The stapedius muscle pulls the stapes ……:

- **a.** Laterally.
- **b.** Medially.
- **c.** Inferiorly.
- **d.** Superiorly.
- **e.** Posteriorly.

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Orbital Region and Eyeball

124. The medial wall of the orbit is formed primarily by the ____ bone:

- **a.** Palatine.
- **b.** Maxillary.
- **c.** Temporal.
- **d.** Zygomatic.
- **e.** Ethmoidal.

125. The lacrimal bone participates in the formation of the ____ of the orbit:

- **a.** Roof.
- **b.** Floor.
- **c.** Apex.
- **d.** Medial wall.
- **e.** Lateral wall.

126. The apex of the orbit is in the ___ bone:

- **a.** Orbital part of the frontal.
- **b.** Squamous part of temporal.
- **c.** Lesser wing of the sphenoid.
- **d.** Perpendicular plate of palatine.
- **e.** Frontal process of the zygomatic.

127. The thickest part of the orbital walls is the:

- **a.** Roof.
- **b.** Floor.
- **c.** Apex.
- **d.** Medial wall.
- **e.** Lateral wall.

128. The medial angel of the eye is known as:

- **a.** Tarsus.
- **b.** Canthus.
- **c.** Conjunctival sac.
- **d.** Palpebral fissure.
- **e.** Conjunctival fornix.

129. Post-synaptic parasympathetic innervation to the lacrimal gland is conveyed through ___ nerve:

- **a.** Vidian.
- **b.** Zygomatic.
- **c.** Nasociliary.
- **d.** Deep petrosal.
- **e.** Greater petrosal.

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130. The _____ muscle opens the eye widely:

- **a.** Medial rectus.
- **b.** Lateral rectus.
- **c.** Superior rectus.
- **d.** Superior oblique.
- **e.** Levator palpebrae superioris.

131. The Abducent nerve supplies the _____ muscle:

- **a.** Lateral rectus.
- **b.** Superior tarsal.
- **c.** Superior rectus.
- **d.** Inferior oblique.
- **e.** Superior oblique.

132. The Superior oblique muscle is supplied by the ___ nerve:

- **a.** Frontal.
- **b.** Lacrimal.
- **c.** Abducent.
- **d.** Trochlear.
- **e.** Oculomotor.

133. The medial rectus muscle _____ the eye:

- **a.** Rotates.
- **b.** Adducts.
- **c.** Abducts.
- **d.** Elevates.
- **e.** Depresses.

134. The pre-synaptic parasympathetic root of the ciliary ganglion arises from the _____ nerve:

- **a.** Facial.
- **b.** Frontal.
- **c.** Abducent.
- **d.** Nasociliary.
- **e.** Oculomotor.

135. The anterior transparent part of the outer (fibrous) layer of the eye is named:

- **a.** Sclera.
- **b.** Retina.
- **c.** Cornea.
- **d.** Choroid.
- **e.** Ciliary body.

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136. The "red eye" phenomenon that occurs in photography is due to the flash reflection from the:

- **a.** Iris.
- **b.** Pupil.
- **c.** Retina.
- **d.** Cornea.
- **e.** Choroid.

137. The dilator pupillae muscle receives postsynaptic sympathetic innervation through the _____ nerve:

- **a.** Facial.
- **b.** Frontal.
- **c.** Lacrimal.
- **d.** Oculomotor.
- **e.** Long cilliary.

138. In pupillary light reflex, the efferent signal is carried by the ___ nerve:

- **a.** Optic.
- **b.** Facial.
- **c.** Abducent.
- **d.** Trochlear.
- **e.** Oculomotor.

139. The "blind spot" in the retina refers to the _____:

- **a.** Sclera.
- **b.** Retina.
- **c.** Optic disc.
- **d.** Ciliary body.
- **e.** Macula lutea.

140. The area of the most acute visual resolution is the:

- **a.** Iris.
- **b.** Pupil.
- **c.** Retina.
- **d.** Foveola.
- **e.** Optic disc.

141. Macula lutea is located _____ to the optic disc:

- **a.** Deep.
- **b.** Lateral.
- **c.** Medial.
- **d.** Inferior.
- **e.** Superior.

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142. In the eye, the main site that is responsible for refraction of the light is:

- **a.** Iris.
- **b.** Lens.
- **c.** Cornea.
- **d.** Aqueous humor.
- **e.** Vitreous humor

143. The lens is adjusted for near vision by its increased convexity due to the contraction of the _____ muscle:

- **a.** Ciliary.
- **b.** Superior rectus.
- **c.** Dilator pupillae.
- **d.** Sphincter pupillae.
- **e.** Levator Palpebrae Superioris.

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Section Two: Scenario-Based MCQs

Clinical Case Scenario (1)

Jaafar is a 24 years old university student who was involved in a fight, presented to the emergency room with lacerations on the side of the face over the mandible, the patient was feeling numbness in the lower lip and chin, examination and imaging studies revealed mandibular fracture and dislocation.

Answer the following questions:

144. Which part of the mandible participates in the formation of the temporomandibular joint (TMJ)?

- **a.** Angle.
- **b.** Ramus.
- **c.** Genial tubercle.
- **d.** Symphysis menti.
- **e.** Condyloid process

145. What is the most common type of mandibular dislocation?

- **a.** Lateral.
- **b.** Superior.
- **c.** Anterior.
- **d.** Posterior.
- **e.** Posterio-superior.

146. What is the most common site of fracture in the mandible?

- **a.** Body.
- **b.** Angle.
- **c.** Symphysis menti.
- **d.** Coronoid process.
- **e.** Condyloid process.

147. What is the most commonly injured nerve following mandibular fracture?

- **a.** Buccal.
- **b.** Lingual.
- **c.** Inferior alveolar.
- **d.** Transverse cervical.
- **e.** Marginal mandibular.

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Clinical Case Scenario (2)

Hashim is a 65 years manual worker, he is brought to the renal unit for hemodialysis, and the doctor plans to perform central venous access with a catheter.

Answer the following questions:

148. What is the best vein for central venous access?

- **a.** Subclavian.
- **b.** Internal jugular.
- **c.** External jugular.
- **d.** Anterior jugular.
- **e.** Brachiocephalic.

149. The common carotid artery is _____ to the internal jugular vein:

- **a.** Medial.
- **b.** Lateral.
- **c.** Anterior.
- **d.** Posterior.
- **e.** Superficial.

150. The internal jugular vein is accessible for puncture or catheterization at the:

- **a.** Jugular notch.
- **b.** Occipital triangle.
- **c.** Omotracheal triangle
- **d.** Posterior cervical region.
- **e.** Lesser supraclavicular fossa.

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Clinical Case Scenario (3)

In preparation of neck dissection (surgical procedure to remove cervical lymph nodes due to metastases), the surgeon gave the junior doctor a task of describing the lymphatic drainage of the head and neck, he asked him the following questions:

Answer the following questions:

151. The lymph from posterior part of the SCALP drains into the ___group of lymph nodes:

- **a.** Buccal.
- **b.** Occipital.
- **c.** Laryngeal.
- **d.** Paratracheal.
- **e.** Anterior cervical.

152. The submental group of lymph nodes drain the lymph from:

- **a.** Upper lib.
- **b.** Parotid gland.
- **c.** Upper incisors
- **d.** Tip of the tongue.
- **e.** External auditory meatus.

153. The superficial cervical lymph nodes are located alongside the___ vein:

- **a.** Subclavian.
- **b.** Internal jugular.
- **c.** External jugular.
- **d.** Anterior jugular.
- **e.** Brachiocephalic.

154. The carotid triangle hosts the ____ group of lymph nodes:

- **a.** Parotid.
- **b.** Mastoid.
- **c.** Laryngeal.
- **d.** Deep cervical.
- **e.** Submandibular.

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Section Three: Extended Matching Questions (R-Type items)

Skull foramina

- **Options: (foramina)**
  - **a.** Optic canal.
  - **b.** Foramen ovale.
  - **c.** Jugular foramen.
  - **d.** Foramen magnum.
  - **e.** Foramen spinosum.
  - **f.** Stylomastoid foramen.
  - **g.** Superior orbital fissure.
- **For each structure mentioned below, select the correct foramen from the options (above) that allows passage to the structure, some of the options (foramina) may not be used and some of them may be used more than once.**

155. Optic nerve.
156. Facial nerve.
157. Trochlear nerve.
158. Vertebral artery.
159. Ophthalmic vein.
160. Ophthalmic artery.
161. Mandibular nerve.
162. Ophthalmic nerve.
163. Oculomotor nerve.
164. Middle meningeal artery.
165. Glossopharyngeal nerve.
166. Accessory meningeal artery.

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Triangles of the Neck

- **Options: (Triangles)**
  - **a.** Carotid.
  - **b.** Occipital.
  - **c.** Muscular.
  - **d.** Submental.
  - **e.** Submandibular.
  - **f.** Supraclavicular.
- **For each structure of the neck mentioned below, select the correct triangle from the options (above) that contains that structure, some of the options (Triangles) may be used more than once.**

167. Facial vein.
168. Vagus nerve.
169. Accessory nerve.
170. Subclavian artery.
171. Thyrohyoid muscle.
172. Sternohyoid muscle.
173. Internal jugular vein.
174. Anterior jugular vein.

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Motor Innervation of the Head and Neck

- **Options: (nerves)**
  - **a.** Facial.
  - **b.** Abducent.
  - **c.** Trochlear.
  - **d.** Mandibular.
  - **e.** Oculomotor.
  - **f.** Hypoglossal.
  - **g.** Inferior alveolar.
  - **h.** Glossopharyngeal.
  - **i.** External laryngeal.
  - **j.** Recurrent laryngeal.
- **For each muscle mentioned below, select the correct nerve from the options (above) that supplies that muscle, some of the options (nerves) may not be used and some of them may be used more than once.**

175. Mylohyoid.
176. Stylohyoid.
177. Geniohyoid.
178. Orbicularis Oris.
179. Occipitofrontalis.
180. Medial Pterygoid.
181. Tensor Veli Palatine.
182. Stylopharyngeus.
183. Vocalis.
184. Posterior cricoarytenoid.
185. Stapedius.
186. Tensor tympani.
187. Lateral rectus.
188. Superior oblique.
189. Medial rectus.
190. Cricothyroid

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Connections to the nasal cavity

- **Options: (nerves)**
  - **a.** Middle meatus.
  - **b.** Inferior meatus.
  - **c.** Superior meatus.
  - **d.** Spheno-ethmoidal recess.
- **For each given nasal sinus (below), select the correct area of the nasal cavity from the options (above) into which the given sinus opens, some of the options (above) may not be used and some of them may be used more than once.**

191. Frontal sinus.
192. Maxillary sinus
193. Sphenoidal sinus.
194. Posterior ethmoidal sinus.
195. Middle ethmoidal air cells.
196. Anterior ethmoidal air cells.

Functions of the Ear

- **Options: (nerves)**
  - **a.** Auricle.
  - **b.** Cochlea.
  - **c.** Stapedius.
  - **d.** Tensor tympani
  - **e.** Pharyngotympanic tube.
- **For each function of the ear mentioned (below), select the correct part of the ear from the options (above) that performs the function, some of the options (above) may not be used and some of them may be used more than once.**

197. Sound waves collection.
198. Inhibition of excessive movement of Stapes.
199. Reduction of the amplitude of Malleus oscillations
200. Equalization of the pressure on both sides of the eardrum.

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Section Four: Medical Imaging

Figure (1): Lateral Cervical Radiograph



Figure (2): Anteroposterior Cervical Radiograph

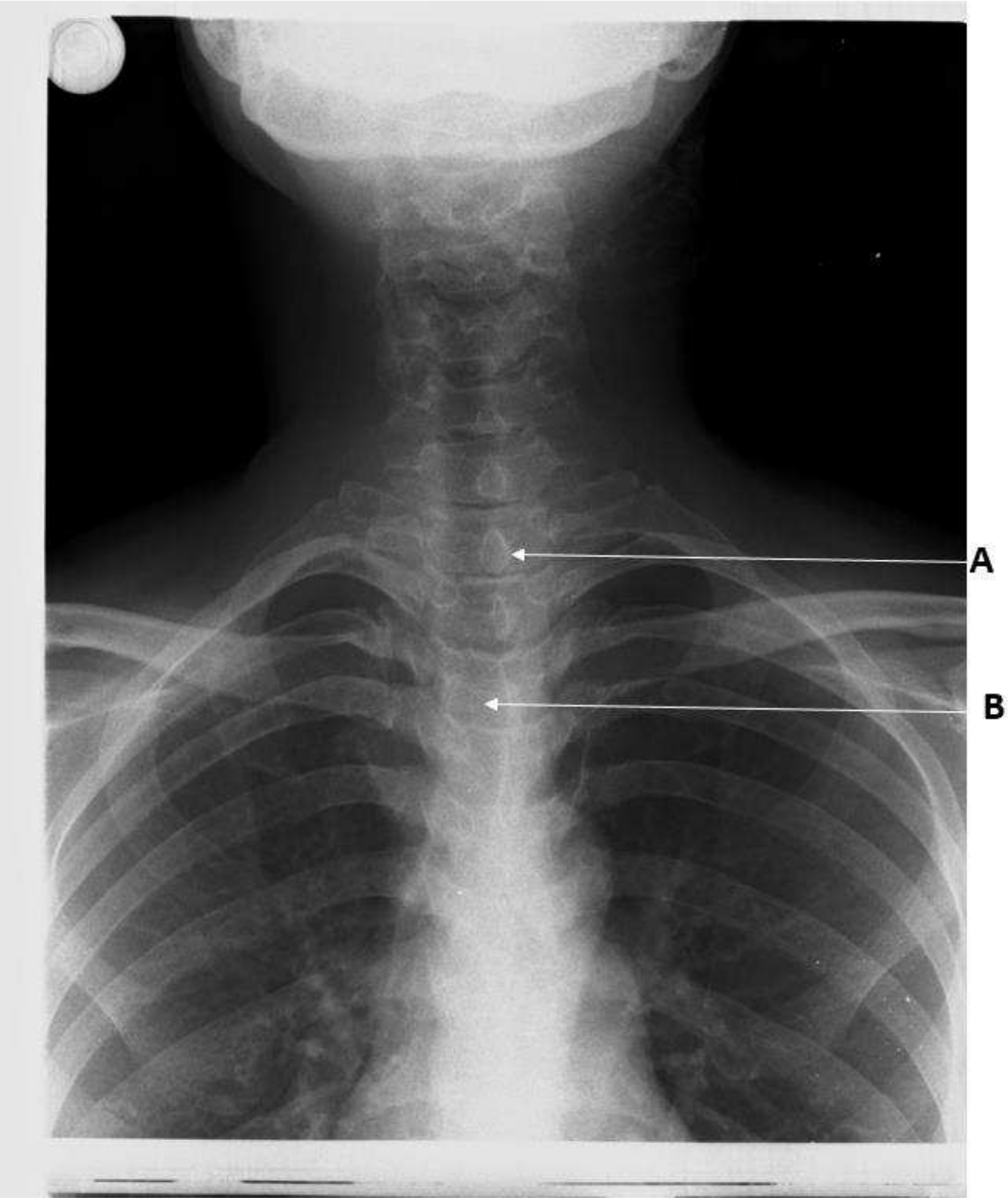
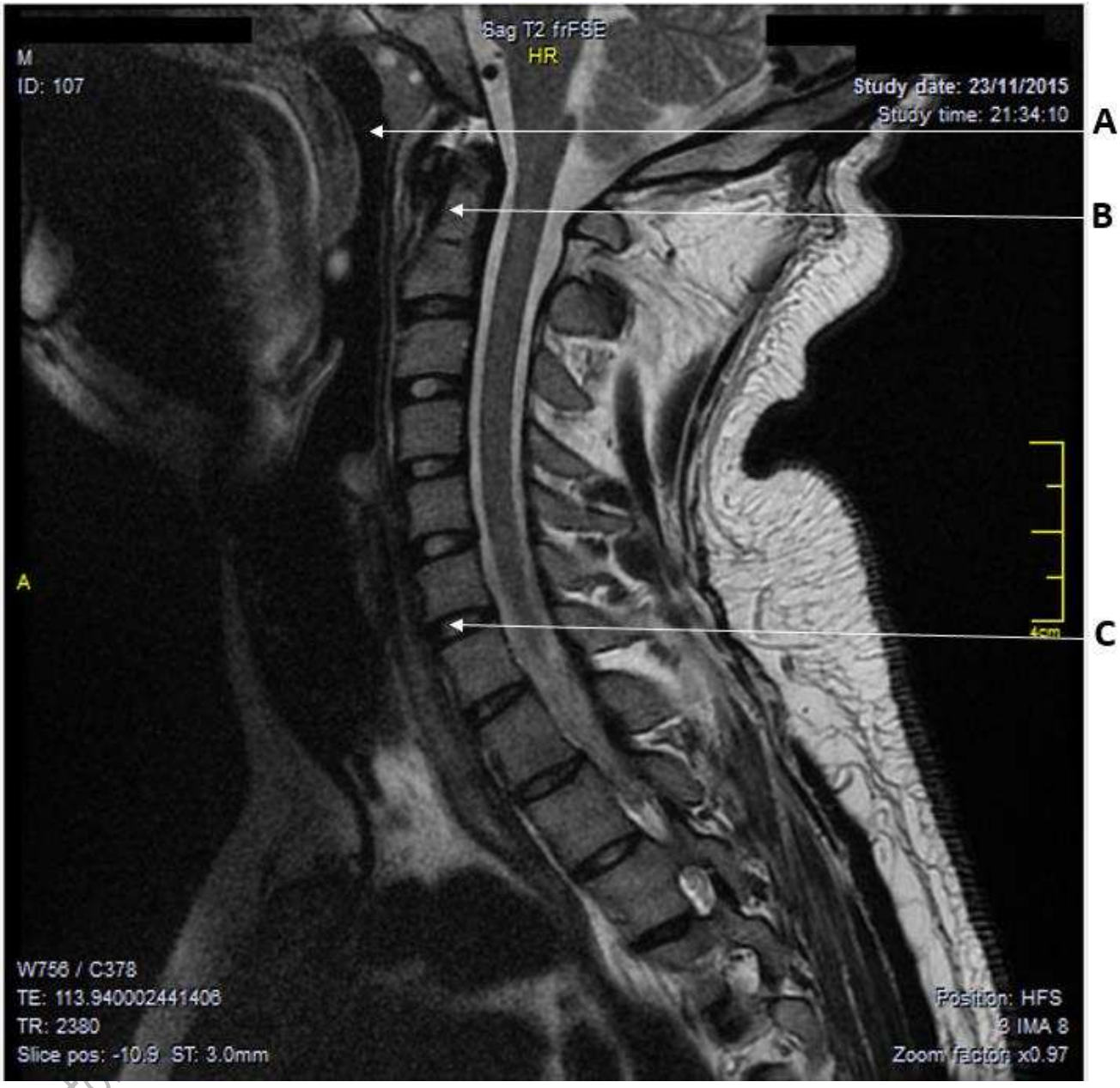


Figure (3): MRI Cervical Spine

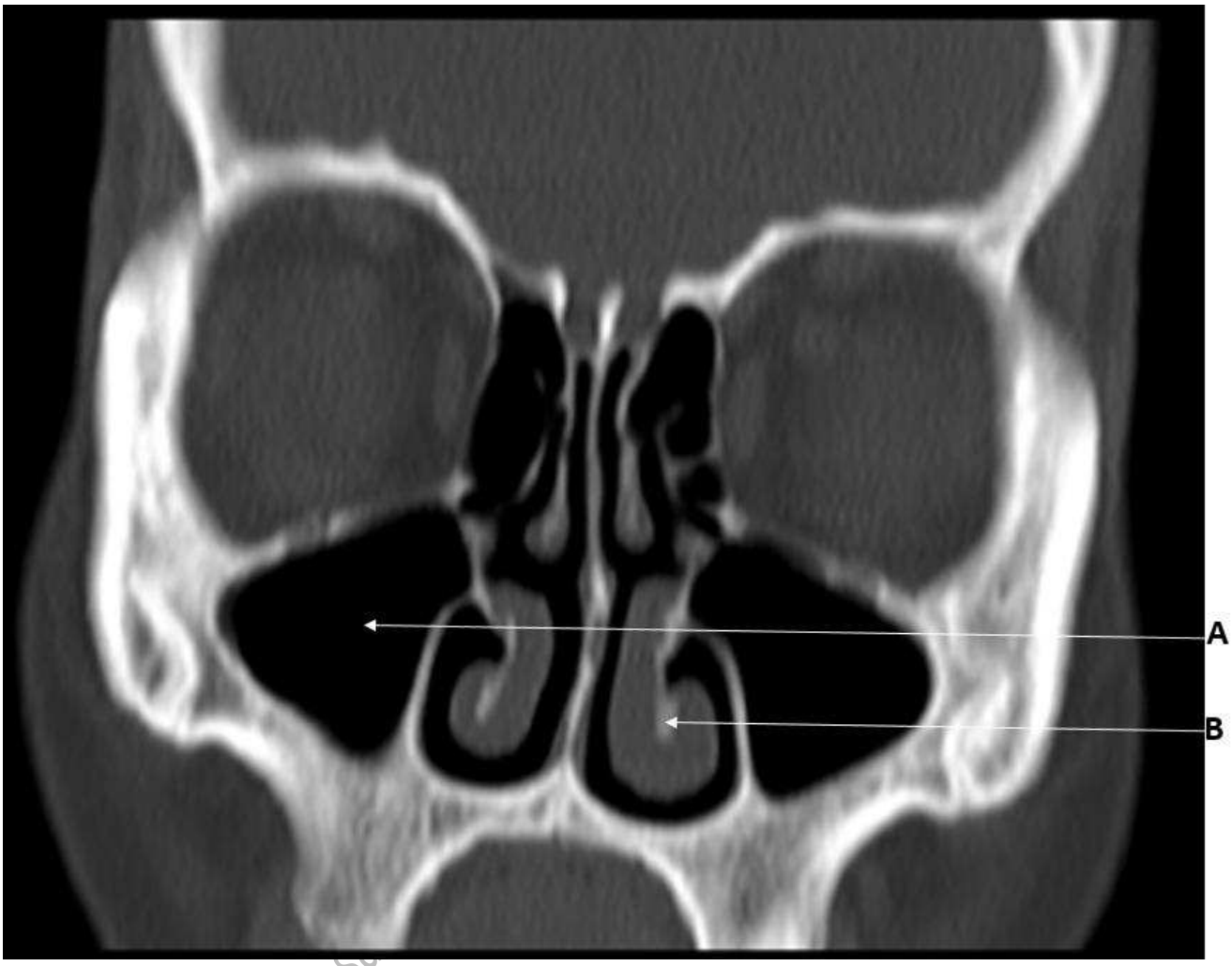


Anato

Figure (4): CT Brain - Bone Window
At the level of petrous part of temporal bone



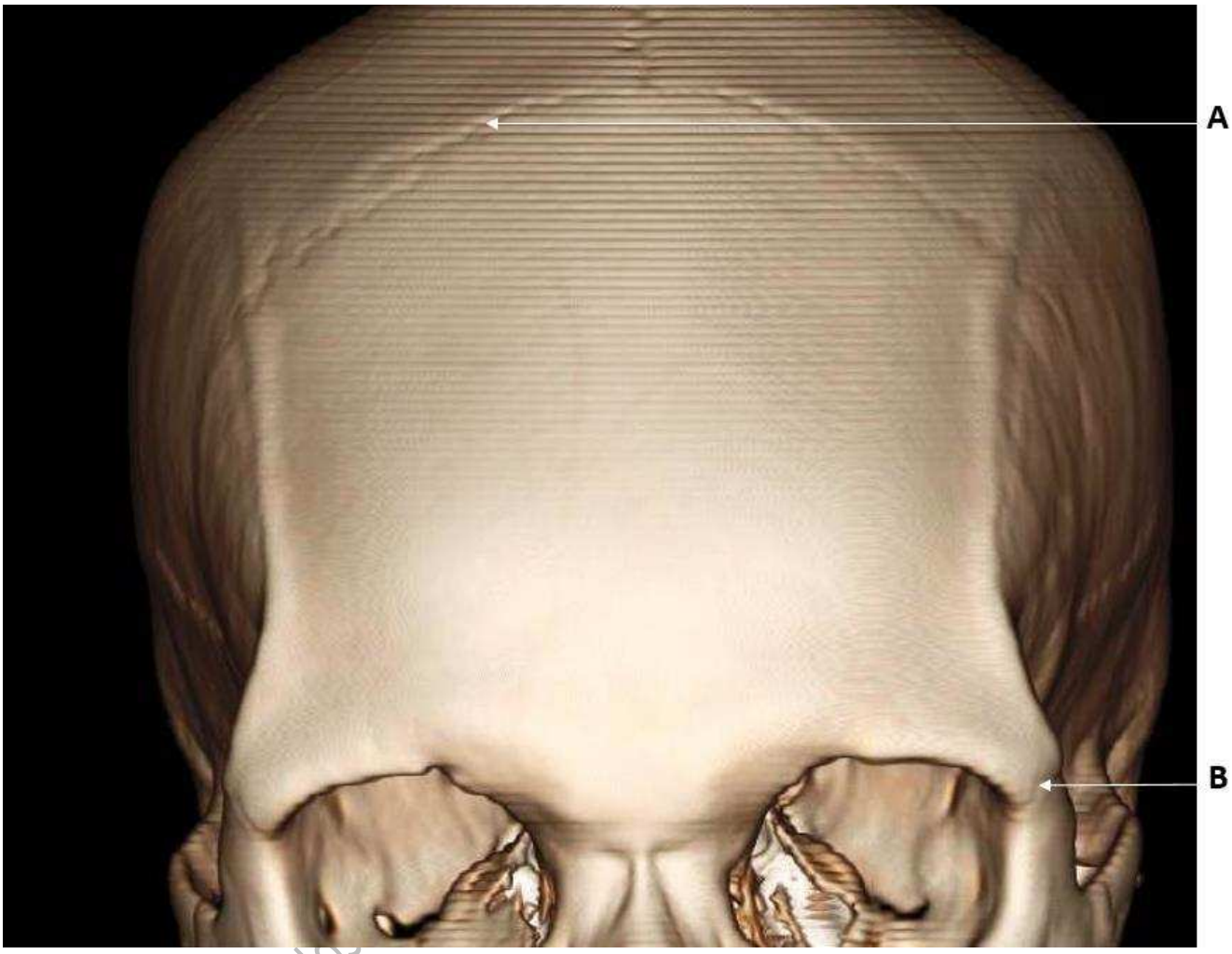
Figure (5): CT Paranasal Sinuses
Coronal view



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Figure (6): 3D reconstruction of CT-Skull

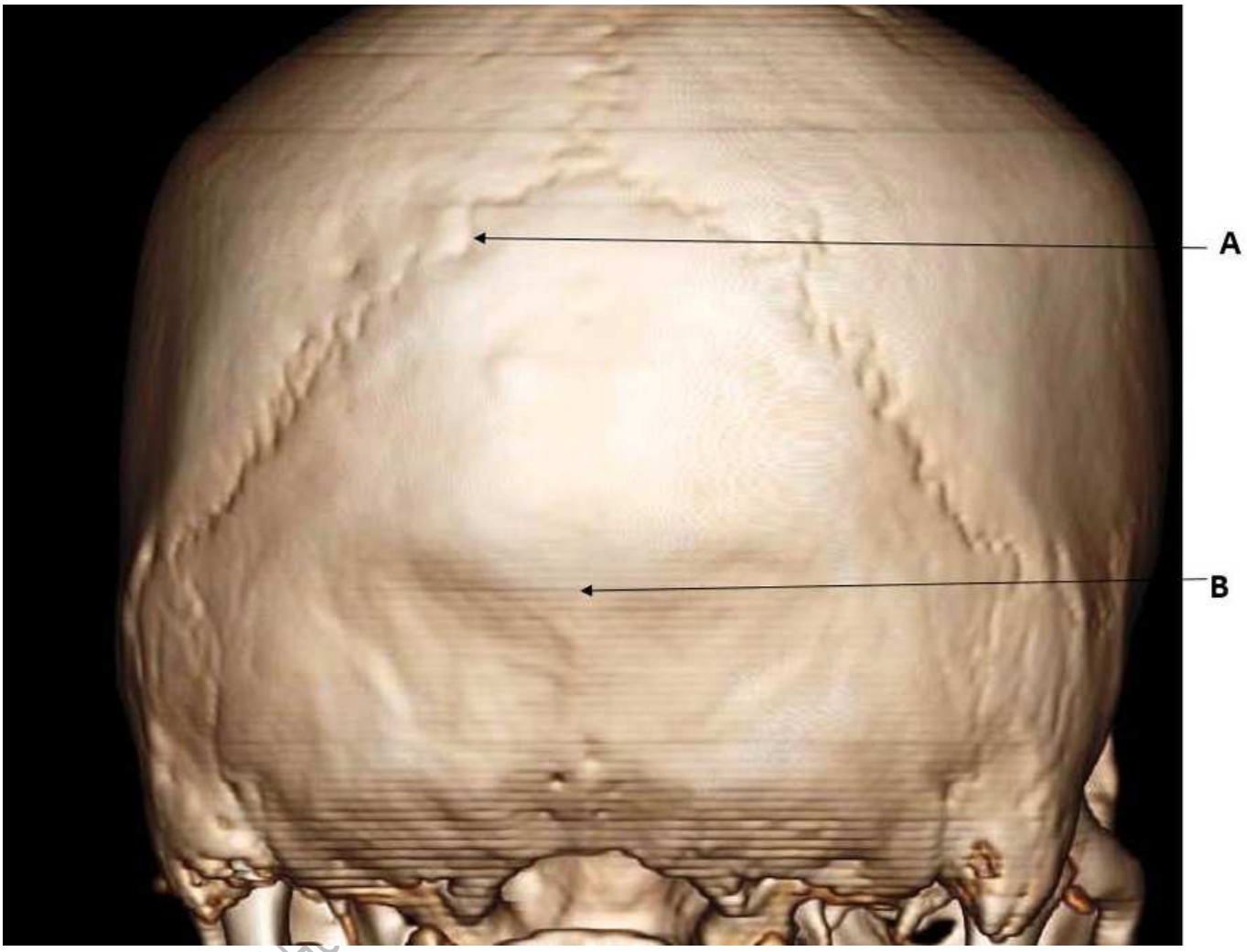
Skull and facial bones



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Figure (7): 3D reconstruction of CT-Skull

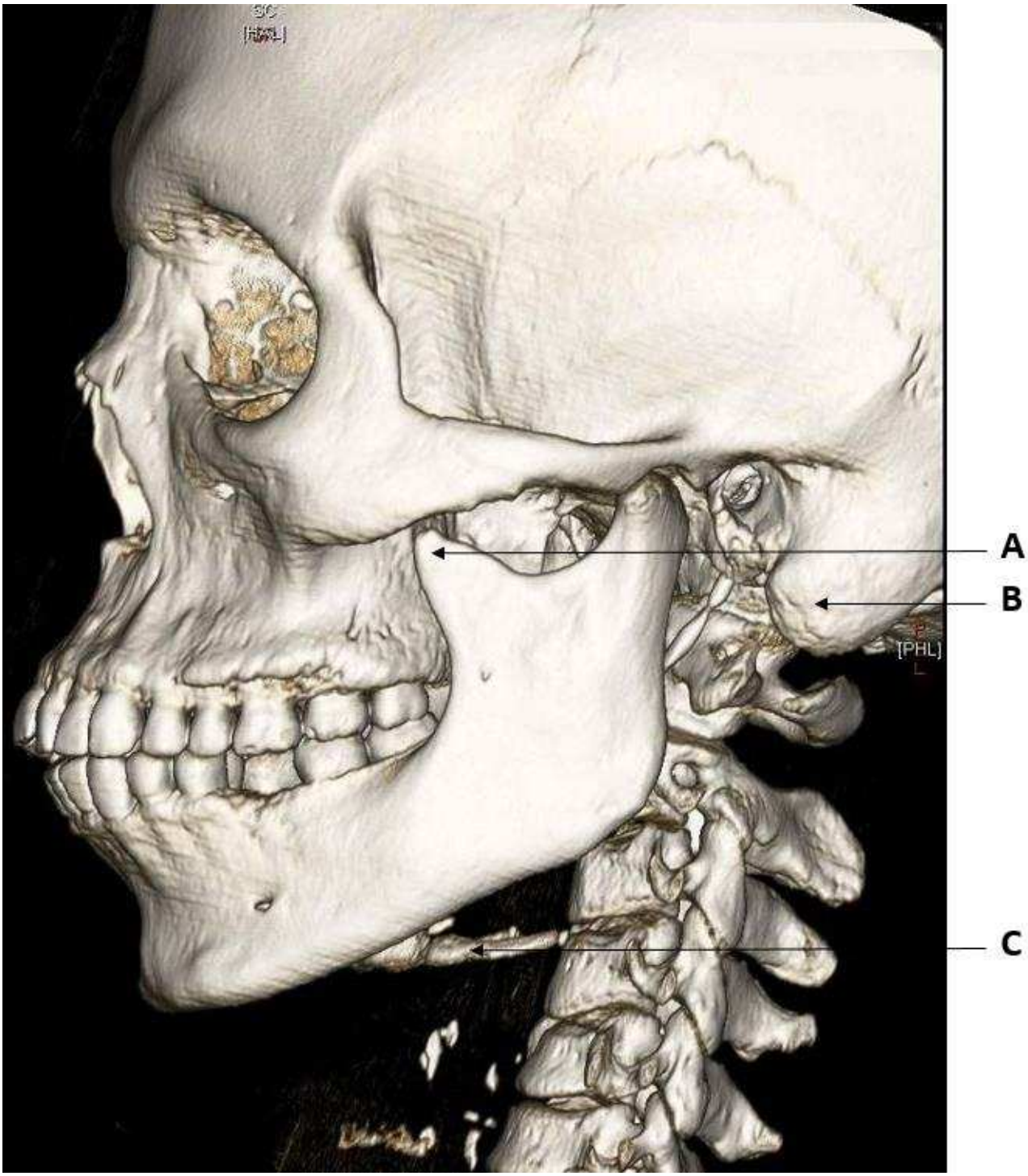
Skull: posterior view



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Figure (8): 3D reconstruction of CT-Skull

Skull: lateral view



On the previous pages there are seven radiological images, all representing normal appearance of the structures in the head and neck, study them carefully and answer the following questions:

Figure (1): Lateral Cervical Radiograph:

This is a lateral cervical radiograph, identify the following structures:

201. This soft tissue structure (A).

202. This bony structure (B).

Figure (2): Anteroposterior Cervical Radiograph:

This is anteroposterior cervical radiograph, identify the following structures:

203. This bony projection (A).

204. This soft structure (B) "hint: air-filled".

Figure (3): MRI Cervical Spine:

This is MRI cervical spine (sagittal view), identify the following structures:

205. This structure (A).

206. This bony structure (B).

207. This structure (C).

Figure (4): CT Brain - Bone Window:

This is CT brain - bone window, identify the following structures:

208. This bony structure (A).

209. This bony structure (B).

210. This bony structure (C).

Figure (5): CT PNS (Paranasal Sinuses), Coronal view

This is a CT PNS (Paranasal Sinuses), Coronal view, identify the following structures:

211. This air-containing space (A)

212. This bony structure (B)

Figure (6): 3D reconstruction of CT-Skull: Skull and facial bones

This is a 3D Reconstruction: Skull and facial bones, identify the following sutures:

213. (A)

214. (B)

Figure (7): 3D reconstruction of CT-Skull:

This is a 3D Reconstruction: Skull: posterior view, identify the following structures:

215. Suture (A).

216. Bony projection (B).

Figure (8): 3D reconstruction of CT-Skull:

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This is a 3D Reconstruction: Skull: lateral view, identify the following sutures structures:

217. Bony structure (A).

218. Bony structure (B).

219. This bone (C).

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Answers

Section One (MCQs Type A)

Skull and cervical vertebrae

1. **Questions (1), (2), (3), (4), (5) and (6):** The main theme of these questions is the facial skeleton (bones of the face), the frontal bone is a single bone consists of three parts: squamous, orbital and nasal parts, it forms a large portion of the anterior aspect of the skull, the superior orbital margin and the floor of the anterior cranial fossa (by the orbital plate), it articulates posteriorly (by its squamous part) with parietal bone and the greater wing of sphenoid (in which it articulates at the Pterion), inferiorly it articulates (by its nasal part) with the nasal bones (inferiorly) and the maxilla and lacrimal bones (inferior and lateral), the orbital part articulates posteriorly with the sphenoid bones, it is divided into two halves by the metopic suture, which disappears at about 8 years. The strongest bone among the bones of the face is the mandible.
2. **See answer (1) above.**
3. **See answer (1) above.**
4. **See answer (1) above.**
5. **See answer (1) above.**
6. **See answer (1) above.**
7. **Questions (7) and (8):** the ethmoid bone contributes to the: floor of the anterior cranial fossa by its cribriform plate, to the medial wall of the orbital cavity, nasal septum and parts of the roof and lateral wall of the nasal cavity.
8. **See answer (7) above.**
9. The optic canal transmits the optic nerve and the ophthalmic artery, the rest of the mentioned arteries in the questions are branches from the maxillary artery in the infratemporal fossa.
10. The **Inion** is a protrusion from the external occipital protuberance, **Pterion** is formed by four bones: frontal ,sphenoid, temporal, and parietal, it is the area of the **anterolateral fontanelle**, the **Bregma** is where the coronal and sagittal sutures meet, it is the area of the **anterior fontanelle**, the **Lambda** is the junction of the lambdoid and sagittal sutures, it is the area of the **posterior fontanelle**, the **Asterion** is point in which parietomastoid, occipitomastoid, and lambdoid sutures meet together, it is the area of the **posterolateral fontanelle.**
11. Ossification of the parietal bone starts by ossification centers that appear at about the seventh week of intrauterine life at the area of the eminence, the

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ossification process (membranous) spreads from the center to the peripheries, the four angels of the bone ossify lastly, contribute to the appearance of the fontanelles.

12. **Questions (12) and (13):** The hypoglossal nerve (which supplies the tongue) emerges from the skull through the hypoglossal canal. The hypoglossal canal, condylar canal, mastoid foramen and foramen magnum are all located in the posterior cranial fossa as part of occipital bone, the only exception is the jugular foramen which is located in-between the occipital bone and the petrous part of temporal bone.
13. **See answer (12) above.**
14. **Questions (14) and (15):** The temporal bone consists of: squamous, petrous, mastoid, tympanic parts and the styloid process. The facial and vestibulocochlear nerves emerge from the posterior cranial fossa through the internal acoustic meatus, which is situated in the medial surface of the petrous part of temporal bone, this part of temporal bone also houses the acoustic labyrinth.
15. **See answer (14) above.**
16. **Questions (16), (17), (18), (19), (20), (21) and (22):** The sphenoid bone consists of body, wings (lesser and greater) and pterygoid plates (medial and lateral). The superior orbital fissure is a gap between the lesser and greater wings. The foramina: rotundum, ovale and spinosum are all located in the middle cranial fossa as a part of the greater wing of sphenoid bone, foramen rotundum transmits the Maxillary division of the trigeminal nerve, foramen ovale transmits the Mandibular division of the trigeminal nerve and lesser petrosal nerve; and foramen spinosum transmits the Middle meningeal artery. The jugular foramen is located between the occipital bone and the petrous part of temporal bone. Foramen lacerum is located between the sphenoid bone and the petrous part of temporal bone and transmits the internal carotid artery. The part of sphenoid bone that contributes to the floor of the middle cranial fossa ossifies in cartilage, the rest of the bone undergoes membranous ossification.
17. **See answer (16) above.**
18. **See answer (16) above.**
19. **See answer (16) above.**
20. **See answer (16) above.**
21. **See answer (16) above.**
22. **See answer (16) above.**

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23. Premature closure of the sagittal suture leads to formation of a long, narrow, wedge-shaped cranium (Scaphocephaly), premature closure of the coronal or the lambdoid suture on one side only leads to the formation of twisted and asymmetrical skull (Plagiocephaly), tower-like cranium (Oxycephaly) results from premature closure of the coronal suture.

24. Questions (24), (25) and (26): the atypical cervical vertebrae are the first (Atlas), the second (Axis) and the seventh.

The Atlas (first cervical vertebra) is atypical because:

It does not possess a body or a spinous process.

It has:

Anterior and posterior arches.

A lateral mass on each side.

The Axis (second cervical vertebra) is atypical because:

It has an upward projection form its body called the odontoid process.

The seventh cervical vertebra is atypical because:

The spine is the longest among all cervical vertebrae, and it is not bifid.

The foramen transversarium does not transmit the vertebral artery.

25. See answer (24) above.

26. See answer (24) above.

27. The articulation sites in the atlanto-axial joints are three: two lateral and one median.

The lateral articulations are between the lateral masses of C1 and C2 (gliding synovial joint).

The median articulation is between the odontoid process of C2 and the anterior arch of the C1 (pivot synovial joint).

The axis passes throught the odontoid process, this joint moves the head from side to side (the "NO" movement)

28. The hyoid bone serves as muscle attachment for supra- and infra-hyoid muscles, these muscles elevate and depress the larynx in response to different phases of swallowing.

Suprahyoid muscle:

Mylohyoid, **geniohyoid** and **digastric** muscles connect the hyoid bone to the mandible, **Stylohyoid** muscle connects the hyoid bone to the styloid process of the temporal bone.

Infrahyoid muscles:

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Omohyoid: connects the hyoid bone to the scapula.
Sternothyroid: connects the sternum to the thyroid cartilage.
Thyrohyoid: connects the thyroid cartilage to the hyoid bone.
Sternohyoid: connects the sternum to the hyoid bone.

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Neck

29. **Questions (29) and (30):** The Platysma muscle is contained within the subcutaneous tissue, attached inferiorly to the fascia over pectoralis major and deltoid muscles, upper attachment is to the lower border of mandible and skin. It is innervated by the facial nerve, it tracks the angels of the mouth downwards as expression of sadness, and the external jugular vein runs deep to it.
30. **See answer (29) above.**
31. **Questions (31) and (32):** The investing layer of deep cervical fascia encloses the submandibular and parotid glands, trapezius and sternocleidomastoid muscles, and thickened to form the Stylomandibular ligament.
32. **See answer (31) above.**
33. The ligament is derived from the pretracheal fascia.
34. The only neck ligament that prevents hyperextension of the neck is the anterior longitudinal ligament.
35. The phrenic nerve runs deep to the prevertebral fascia.
36. The retropharyngeal space extends into the mediastinum.
37. **Questions (37) and (38):** The neck is divided into anterior and posterior divisions using sternocleidomastoid muscle as land mark, it has multiple actions depending on the features of contractions:

    Unilateral contraction: flexes the neck laterally (Moving the head so that the face looks upwards and to the opposite side).

    Bilateral contraction: extends the neck at the atlanto-occipital joints.

    Assists in respiration: elevates the manubrium when the cervical vertebrae are fixed.

    The transverse cervical nerve passes over it superficially anteriorly in a horizontal course,
38. **See answer (37) above.**
39. The digastric "submandibular" triangle is bounded by the anterior and posterior bellies of digastric muscle and the lower border of the mandible.
40. The posterior triangle is divided into two triangles: occipital and supraclavicular, by Omohyoid muscle.
41. One of the contents of the submandibular triangle is the hypoglossal nerve which innervates the muscles of the tongue.
42. **Questions (42), (43), (44), (45) (46), (47), (48), (49) and (50):** the **cervical plexus** arises from C1 to C4 (anterior rami), roots pass between Scalenus Anterior and Medius, the cutaneous branches of the cervical plexus emerge at

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the middle of the posterior border of Sternocleidomastoid muscle **"nerve point of the neck"**, the transvers cervical nerve crosses over the muscle and passes horizontally to the midline of the neck.

Ansa cervicalis "Ansa hypoglossi" is a nerve loop formed by union of two roots: superior and inferior, the **superior root** arises from C1, the **inferior root** arises from the fused C2 and C3. It runs downwards on the lateral aspect of the internal jugular vein.

The **cutaneous branches** of cervical plexus are:

Lesser occipital nerve: supplies the skin posterior and superior to the auricle (back of the SCALP)

Great auricular nerve: supplies the skin over the mastoid process and over the mandible at its angel, it also supplies the fascia of the parotid gland.

Supraclavicular nerve: supplies the skin over the shoulder.

Transverse cervical nerve: supplies the skin of the neck (anteriorly).

The **muscular branches** supply the prevertebral muscles and some of the muscles of the neck, namely: Omohyoid, Sternohyoid, and Sternothyroid muscles. The Thyrohyoid muscle is supplied by a branch that emerges from the hypoglossal nerve, this branch is C1 actually.

The **phrenic nerve** is muscular to the diaphragm and sensory to parts of pleura, pericardium and peritoneum.

43. See answer (42) above.

44. See answer (42) above.

45. See answer (42) above.

46. See answer (42) above.

47. See answer (42) above.

48. See answer (42) above.

49. See answer (42) above.

50. See answer (42) above.

51. Questions (51) and (52): The thyroid gland, consisting of right and left lobes and isthmus, lies in the neck anterior to the trachea at the level of the C5 to T1 vertebrae, deep to the sternothyroid and sternohyoid muscles. The isthmus is positioned anterior to the second and third tracheal rings.

It moves with swallowing because it is tight by the pretracheal layer of deep cervical fascia.

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The superior laryngeal artery accompanies the internal branch of the superior laryngeal nerve, while the inferior laryngeal artery runs close to the recurrent laryngeal nerve.

52. See answer (51) above.

53. Questions (53) and (54): Except for the cervical ganglion, all the mentioned ganglia in the question are parasympathetic, injury to the cervical ganglion (sympathetic) will lead to the following consequences of **Horner's syndrome**:

Miosis: pupillary constriction, due to paralysis of the sympathetically innervated dilator pupillae muscle.

Ptosis: drop of the upper eyelid, due to paralysis of the sympathetically innervated part of the levator palpebrae superioris muscle.

Enophthalmos: Sinking in of the eyeball (the eye is displaced backwards)

Anhydrosis: Vasodilation and absence of thermal sweating on the face and neck.

Johann Friedrich Horner (1831–1886) is a Professor of Ophthalmology in the University of Zurich, Switzerland.

54. See answer (53) above.

55. Xerophthalmia refers to dryness of the eye, it is associated with vitamin A deficiency,

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Face and SCALP

56. Questions (56), (57), and (58): The branches of the first division (ophthalmic) of the trigeminal nerve are:

Nasociliary:

Posterior ethmoidal.

Anterior ethmoidal (continues as **external nasal**): supplies outer (dorsal) surface of the nose and the apex (tip) of the nose.

Infratrochlear: to the skin just lateral to the root of the nose; skin and conjunctiva of medial aspects of the eyelids.

Frontal: supplies the forehead and SCALP.

Supraorbital: to the skin and conjunctiva of the upper eyelid; forehead and anterior part of the SCALP.

Supratrochlear: the same as above, it is located medial to the supraorbital nerve.

Lacrimal: to the lacrimal gland and part of skin and conjunctiva of the upper eyelid.

57. See answer (56) above.

58. See answer (56) above.

59. The maxillary nerve supplies the skin of the face by three nerves: zygomaticofacial, Zygomaticotemporal and infraorbital, the later is a direct continuations of the maxillary nerve.

60. Questions (60), (61) and (62): Muscle of the following facial expressions are:

Compressing cheek against molar teeth: Buccinator muscle.

Smile: zygomaticus major and levator anguli oris.

Surprise: frontal belly of Occipitofrontalis muscle.

61. See answer (60) above.

62. See answer (60) above.

63. Questions (63), (64) and (65): Innervation of the parotid gland:

Type of innervation	From	Through			Distribution
Sensory	Cervical plexus	Great auricular (C2 and C3)			parotid sheath and the overlying skin
	Mandibular (from trigeminal nerve)	Auriculotemporal nerve			Parotid gland
Parasympathetic	Inferior salivary nucleus	Glossopharyngeal nerve (presynaptic)	Otic ganglion	Auriculotemporal nerve (postsynaptic)	
Sympathetic	Cervical ganglia	External carotid nerve plexus			

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64. See answer (63) above.

65. See answer (63) above.

66. The otic ganglion is parasympathetic ganglion:

Roots:

Parasympathetic: from inferior salivary nucleus through the glossopharyngeal nerve, its tympatic branch, tympatic plexus, lesser petrosal nerve.

Sympathetic: from the superior cervical ganglion through middle meningeal artery associated fibers.

Sensory: from cell bodies in the trigeminal ganglion through auriculotemporal nerve.

Branches:

To the parotid gland via auriculotemporal nerve.

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Temporal, infratemporal and pterygopalatine fossae

67. Questions (67), (68), (69) and (70): Muscles of mastication:

Muscle	Origin	Insertion	Action on TMJ
Temporalis	Temporal fossa	Coronoid process and anterior aspect of ramus of mandible	Elevation and retraction of the mandible
Masseter	Maxillary process of zygomatic bone and the zygomatic arch	ramus of mandible : angle and lateral surface	Elevation of the mandible
Medial pterygoid	Medial surface of lateral pterygoid plate	Medial surface of ramus of mandible	Elevation, forward and medial deviation
Lateral pterygoid	Lateral surface of lateral pterygoid plate	Anteromedial aspect of neck of mandible	Bilateral: protraction and depression Unilateral: moving jaw to the contralateral side

All the muscles of mastication are supplied by branches from the mandibular nerve.

68. See answer (67) above.

69. See answer (67) above.

70. See answer (67) above.

71. Questions (71) and (72): The mandibular nerve:

Division	Branch	Distribution
Main trunk	Nerve to the medial pterygoid	Medial pterygoid muscle
		Tensor veli palatini muscle
Anterior division (all branches are motor except the Buccal nerve)	Masseteric nerve	Masseter muscle
	Deep temporal nerves	Temporalis muscle
	Nerve to lateral pterygoid	Lateral pterygoid muscle
	Sensory: Buccal	Skin and the mucous membrane of the cheek.
Posterior division (all branches are sensory except the Mylohyoid nerve)	Auriculotemporal nerve	Skin of the auricle, external auditory meatus, TMJ and the scalp.
	Lingual nerve	General and special sensation from the anterior two thirds of the tongue and preganglionic para-sympathetic secretomotor fibers to the submandibular ganglion.
	Inferior alveolar nerve	Teeth, mandible and the skin over the mandible (chin).
	Motor: Mylohyoid nerve (branch from the Inferior alveolar nerve)	Mylohyoid muscle and the anterior belly of the digastric muscle.

72. See answer (71) above.

73. The pterygopalatine fossa is bounded:

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Posteriorly: by the pterygoid process of the sphenoid.
Anteriorly: by the posterior aspect of the maxilla.
Medially: by the perpendicular plate of the palatine bone.
Superiorly: by the greater wing of the sphenoid.

74. Questions (74) and (75): The maxillary nerve enters the pterygopalatine fossa through the foramen rotundum and leaves it through the inferior orbital fissure to get into the orbital cavity, its continuation is the infraorbital nerve.

In the fossa it gives ganglionic, zygomatic and superior alveolar branches (anterior middle posterior).

It is connected to the lacrimal branch of the ophthalmic nerve through its zygomatico-temporal nerve, a branch from the zygomatic nerve.

75. See answer (74) above.

76. The pterygopalatine ganglion is a parasympathetic ganglion. It is attached to the maxillary nerve in the pterygopalatine fossa.

Roots:

Sensory: from the trigeminal ganglion through branches of maxillary nerve.

Parasympathetic: **From**: the superior salivatory nucleus. **Through**: the facial nerve.**Then**: greater petrosal nerve.

Sympathetic: **From**: the superior cervical ganglion. **Through**: deep petrosal nerve.

The deep petrosal nerve joins greater petrosal nerve to form the nerve of the pterygoid canal (Vidian nerve).

Branches: Orbital, nasal, pharyngeal and palatine branches.

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Oral Cavity and palate

77. Questions (77) and (78): The floor of the oral cavity is formed by the mylohyoid muscle it is attached to the mylohyoid line in the medial surface of the body of mandible laterally, and to the median fibrous raphe and the body of hyoid bone at the midline.

Anatomical relations:

Structures located inferiorly:

The platysma and the anterior belly of digastric muscles.

The superficial part of the submandibular gland.

The facial and submental vessels.

The mylohyoid nerve and vessels.

Structures located superiorly:

Muscles: geniohyoid, part of hyoglossus and styloglossus.

Nerves: hypoglossal and lingual nerves, the submandibular ganglion,

Vessels: lingual and sublingual.

Salivary gland: the sublingual gland, the deep part of the submandibular gland and the submandibular duct.

78. See answer (77) above.

79. The Genioglossus muscle originates from the genial tubercle in the posterior surface of the central part of the body of the mandible (posterior to the mandibular symphysis).

The lower fibers are attached to the hyoid body near the midline.

The muscles of opposite sides are merged together at the midline.

Innervation: hypoglossal nerve.

Action: protrudes the apex of the tongue anteriorly.

Bilateral action: it depresses the tongue.

Unilateral action: the tongue deviates to the opposite side.

80. Intrinsic muscles of the tongue:

Muscle	Action
Superior longitudinal	Elevaties apex and sides of tongue. Shortens (retrudes) tongue
Inferior longitudinal	Depresses apex. Shortens (retrudes) tongue
Transverse	Narrows and elongates (protrudes) tongue
Vertical	Flattens and broadens tongue

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81. Sensory innervation of the tongue:

	Innervation for general sensation	Innervation for special sensation (taste)
Anterior two thirds	The lingual nerve, a branch of CN V3	The chorda tympani nerve, a branch of CN VII(except for the vallate papillae).
Posterior one thirds	Glossopharyngeal nerve (CN IX).	Glossopharyngeal nerve (CN IX).

82. All the muscles of the tongue are innervated by the hypoglossal nerve except the palatoglossus (actually a palatine muscle supplied by the pharyngeal plexus).

83. Questions (83) and (84): Functions of muscles during the first (voluntary) stage of swallowing:

Mylohyoid: raises the floor of the mouth in the first stage of swallowing
Tensor veli palatine: strengthens soft palate.
Levator veli palatine: raises soft palate
Palatopharyngeus: moves walls of pharynx upwards, forwards and medially

84. See answer (83) above.

85. All the muscles of the palate are innervated by the pharyngeal plexus except the Tensor veli palatini mascle.

86. Innervation to the teeth and their adjacent alveolar bone:

<table>
<tr><th colspan="2" rowspan="2">Teeth/nerve</th><th colspan="2">Adjacent alveolar bone/nerve</th></tr>
<tr><th>External aspect</th><th>Internal aspect</th></tr>
<tr><td rowspan="4">Upper teeth</td><td>Incisors</td><td colspan="2" rowspan="2">Anterior superior alveolar</td><td rowspan="2">Nasopalatine</td></tr>
<tr><td>Canines</td></tr>
<tr><td>Premolars</td><td colspan="2">Middle superior alveolar</td><td rowspan="2">Greater palatine</td></tr>
<tr><td>Molars</td><td colspan="2">Posterior superior alveolar</td></tr>
<tr><td rowspan="4">Lower teeth</td><td>Incisors</td><td rowspan="4">Inferior alveolar</td><td rowspan="3">Mental</td><td rowspan="4">Lingual</td></tr>
<tr><td>Canines</td></tr>
<tr><td>Premolars</td></tr>
<tr><td>Molars</td><td>Buccal</td></tr>
</table>

87. Questions (87) and (88): Palatine tonsil is located in a tonsillar sinus (fossa), at the oropharynx. The lingual tonsils also located at the oropharynx, at the posterior part of the tongue.

88. See answer (87) above.

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Nasal Cavity and Paranasal Sinuses

89. Questions (89), (90) and (91): Bones that form the boundaries and septum of the nasal cavity:

Roof: frontal, nasal, ethmoidal, and sphenoidal.

Floor: maxilla and palatine bone.

Lateral wall: maxilla, palatine bone, ethmoid, sphenoid and inferior concha (the inferior concha is a separate bone).

Medial wall (nasal septum): vomer and ethmoid.

90. See answer (89) above.

91. See answer (89) above.

92. The connection between the anterior cranial cavity and the roof of the nasal cavity is through the cribriform plate.

93. The Naso-lacrimal Duct opens into the inferior meatus.

94. Questions (94) and (95): The main arterial supply for the nasal cavity is provided by the sphenopalatine artery, however, several arteries contribute to the blood supply, the (Little's area) is located at the lower anterior part of the septum, in which an anastomosis occurs between septal branch of the superior labial, the ascending branch of greater palatine and the sphenopalatine arteries, to form a plexus known as Kieselbach's plexus, this area is a common site of nasal bleeding.

95. See answer (94) above.

96. Questions (96), (97) and (98): Sensory innervation of the nasal cavity is as follows:

<table>
<tr><td colspan="2">Olfactory area (Roof, superior concha and the Upper part of the septum)</td><td colspan="2">Olfactory nerve (Special sensation)</td></tr>
<tr><td rowspan="5">Lateral wall</td><td>Anterior upper</td><td rowspan="5">General sensation</td><td>anterior ethmoidal nerve</td></tr>
<tr><td>Anterior lower</td><td>anterior superior alveolar nerve</td></tr>
<tr><td>Posterior upper</td><td>Posterior superior nasal from the pterygopalatine</td></tr>
<tr><td>Posterior lower</td><td>greater palatine</td></tr>
<tr><td colspan="2">Vestibular area</td><td>Infraorbital nerve</td></tr>
<tr><td rowspan="3">Septum</td><td>Posterior upper</td><td colspan="2">Olfactory nerve (Special sensation)</td></tr>
<tr><td>Anterior</td><td rowspan="2">General sensation</td><td>Anterior ethmoidal and nasopalatine nerves</td></tr>
<tr><td>Posterior</td><td>Medial posterior superior nasal nerve</td></tr>
</table>

97. See answer (96) above.

98. See answer (96) above.

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99. Questions (99) and (100): The anatomical relations of the sphenoidal air sinus are:

Superiorly: the pituitary fossa

Inferiorly: the roof of the nasopharynx.

Laterally: the cavernous sinus and the internal carotid artery.

Posteriorly: the pons.

The sensory innervation of this sinus if provided by the Posterior ethmoidal.

100. See answer (99) above.

101. The boundaries of the **maxillary sinus** are as follows:

Apex: to the zygomatic bone.

Base: lower part of the lateral wall of the nasal cavity.

Roof: floor of the orbital cavity.

102. The nasopharyngeal tonsils (when enlarged are called Adenoid) are located as the name indicates at the nasopharynx.

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Pharynx and Larynx

103. The pharynx is the upper part of the gastrointestinal tract, it extends lower down to the level of the 6th cervical vertebra.

104. The pharynx is divided into three parts: the upper part, nasopharynx (posterior to the nasal cavity), the middle part, oropharynx (posterior to the oral cavity) and the lower part, the laryngopharynx (hypopharynx), located posterior to the larynx.

105. The roof of the nasopharynx is built up by two bones: the basilar part of the occipital bone and the sphenoidal bone.

106. The attachments of the pharyngeal muscles are as follow:

Muscle	Origin	Insertion
Superior constrictor	Pterygoid hamulus, pterygomandibular raphe and the mylohyoid line of mandible (posterior part)	Pharyngeal tubercle
Middle constrictor	Stylohyoid ligament and the hyoid bone (greater and lesser horns)	Pharyngeal raphe
Inferior constrictor	The oblique line of the thyroid cartilage	Posterior to the middle constrictor muscle
Palato-pharyngeus	Hard palate	Lateral sides of the pharynx and esophagus.
Salpingo-pharyngeus	Pharyngotympanic tube	Merges with the Palatopharyngeus
Stylo-pharyngeus	The styloid process of the temporal bone	Thyroid cartilage

107. All the muscles of the pharynx are innervated by the pharyngeal plexus except the stylopharyngeus muscle, which is supplied by the glossopharyngeal nerve.

108. Questions (108) and (109): The thyroid cartilage is one of six cartilages of the larynx: single cartilages:thyroid, cricoid, and epiglottic. Paired cartilages: arytenoid, corniculate, and cuneiform. The upper border of thyroid cartilage lies at a level with C4 Vertebra. The cricoid cartilage lies at a level with C46Vertebra:

109. See answer (108) above.

110. Types of joints of the larynx:

Cricothyroid joint: synovial.
Cricoarytenoid joint: synovial.
Arytenocorniculate joint: synovial or fibrous.

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111. The laryngeal cavity can be divided into the following comparrments:
Vestibule: above the vestibular folds.
Middle part of the cavity: between the vestibular and vocal folds.
Ventricle: recesses found lateral to the middle part of the laryngeal cavity.
Infraglottic cavity: below the vocal folds and above to the cricoid cartilage.

112. The suprahyoid muscles elevate the hyoid bone, the infrahyoid muscles depress the bone.

113. Actions of muscles of the larynx:

Muscle	Main Action(s)
Cricothyroid	Stretches and tenses vocal ligament
Thyroarytenoid	Relaxation of the vocal ligament
Posterior cricoarytenoid	Abduction of the vocal folds
Lateral cricoarytenoid	Adduction of the vocal folds
Transverse and oblique arytenoids	Adduction of the arytenoid cartilages
Vocalis	Increases the vertical length of the vocal folds

114. All the muscles of the larynx are innervated by the recurrent laryngeal nerve except the cricothyroid muscle, which is innervated by the external laryngeal nerve.

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Ear

115. Questions (115) and (116): The **auricle** of the ear is an irregular surface made of elastic cartilage lined externally by skin.

Depressions are: triangular fossa, scaphoid fossa, interagic notch and the concha, which is the deepest depression.

Elevations are: helix, antihelix, tragus and antitragus.

The lobule is made of dens fibrous tissue.

Innervation of the auricle:

Great auricular nerve:

Medial surface of the ear.

Helix, antihelix, and lobule of the lateral surface of the ear.

Lesser occipital nerve:

Upper part of the medial surface of the ear.

Auriculotemporal nerve:

Area anterior to the external acoustic meatus.

Vagus and **Facial** nerves.

116. See answer (115) above.

117. The **external acoustic meatus** is a canal that connects auricle to the tympanic membrane, it is S-shaped.

It is composed of two parts: cartilaginous lateral **third** and bony **medial two-thirds**.

Directions: **pars externa:** medially, anteriorly and slightly up, **pars media:** posteromedially, **pars interna:** anteromedially and slightly down.

118. Questions (118), (119), (120) and (121): walls of the middle ear:

The **anterior (carotid) wall**, features: pharyngotympanic tube's opening and a canal for the tensor tympani.

The **posterior (mastoid) wall**, features: opening to the mastoid antrum.

The **medial (labyrinthine) wall**, features: oval and round windows, the promontory and prominences for facial and lateral semicircular canals.

The **lateral (membranous) wall**: consists mainly of the tympanic membrane.

The **roof (tegmental wall)** is formed by the tympanic part of temporal bone, it is related superiorly to the middle cranial fossa.

The **floor (jugular wall)** is related inferiorly to the internal jugular vein.

119. See answer (118) above.

120. See answer (118) above.

121. See answer (118) above.

122. Questions (122) and (123): Muscles actions and attachments:

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Muscle	Origin	Insertion	Innervation	Action
Tensor tympani	Pharyngotympanic tube (Cartilaginous part) and greater wing of sphenoid	Handle of malleus	Mandibular nerve	Tenses the tympanic membrane by pulling the handle of malleus medially
Stapedius	Posterior wall of the middle ear	Neck of stapes	Facial nerve	Tenses the tympanic membrane by pulling the stapes posteriorly

123. See answer (122) above.

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Eye

124. Questions (124), (125), (126) and (27):

The walls of the orbital cavity:

The **superior wall (roof)** is made of the frontal bone (orbital part), with minor contribution from the sphenoid bone (lesser wing) at the apex of the orbital cavity.

The **inferior wall (floor)** is largely made of the maxillary bone with contributions from the zygomatic and palatine bones.

The **medial wall** is made of the ethmoid bone mainly, and partly with the frontal, lacrimal, and sphenoid bones.

The **lateral wall** is made of the zygomatic (frontal process) and the sphenoid (greater wing) bones. it is the thickest wall.

The apex of the orbit is at the optic canal in the lesser wing of the sphenoid just medial to the superior orbital fissure.

125. See answer (124) above.

126. See answer (124) above.

127. See answer (124) above.

128. These are the definitions of the following terms:

Tarsus: Dens connective tissue that supports the upper and lower eyelids (superior and inferior tarsi)

Canthus: the medial angle of the eye.

Conjunctival sac: it is the space contained by the palpebral conjunctiva and bulbar conjunctiva.

Palpebral conjunctiva: mucous membrane that lines the eyelids internally.

Bulbar conjunctiva: mucous membrane that lines the anterior surface of the eyeball.

Conjunctival fornix: the superior and inferior conjunctival fornices are the spaces bound by the reflections of the palpebral and bulbar conjunctivae when they meet.

Palpebral fissure: the gap between the eyelids.

129. Nerve supply of the lacrimal gland:

Parasympathetic:

Origin: Superior Salivary Nucleus.

Course: (Pre-Synaptic): Facial nerve, then, Greater petrosal nerve, then, Vidian nerve (nerve of the pterygoid canal), then, Pterygopalatine ganglion.

Sympathetic

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Origin: superior cervical ganglion

Course: Internal carotid plexus, then, Deep petrosal nerve, then, Vidian Nerve (the nerve of the pterygoid canal), then, Pterygopalatine ganglion.

From the Pterygopalatine ganglion: both sympathetic and parasympathetic fibers pass within: Zygomatic, then, Zygomaticotemporal, then, lacrimal nerve.

130. Questions (130), (131), (132) and (133): extraocular muscles:

Muscle		Action	Origin	Insertion	Innervation
Orbicularis Oculi	Palpebral part	Lightly closes the eyelids	Medial palpebral ligament	Lateral palpebral raphe	Facial nerve
	Orbital part	Powerfully closes the eyelids	frontal and maxillary bones medial palpebral ligament	Fibers run around the orbit	
Levator palpebrae superioris		Raises upper eyelid	Sphenoid bone	Superior tarsus and the skin of superior eyelid	Oculomotor nerve and sympathetic fibers
Superior oblique		Abduction, depression, and medial rotation of the eyeball	Sphenoid bone	Sclera	Trochlear nerve
Inferior rectus			Common tendinous ring		Oculomotor nerve
Superior rectus		Adduction, elevation and lateral rotation of the eyeball			
Inferior oblique			Floor of orbit		
Medial rectus		Adduction of the eyeball	Common tendinous ring		
Lateral rectus		Abduction of the eyeball			Abducent nerve

131. See answer (130) above.

132. See answer (130) above.

133. See answer (130) above.

134. The roots of the ciliary ganglion:

Sensory fibers from the ophthalmic nerve through the nasociliary nerve.

Parasympathetic fibers from the oculomotor nerve.

Sympathetic fibers from the superior cervical ganglion.

135. Questions (135) and (136):

The eye ball is made by three layers:

Outer (Fibrous) layer:

Sclera (opaque layer) and cornea (the anterior transparent part).

Middle (Vascular) layer:

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The choroid (red eye reflection), ciliary body, and iris.

Inner layer: the retina

136. See answer (135) above.

137. Regarding the muscles of the iris:

Muscle	Sphincter pupillae muscle	Dilator pupillae muscle
Fibers arrangement	Circular	Radial
Innervation	Parasympathetic (Oculomotor nerve)	Sympathetic (long ciliary nerve)
Function	Constricts the pupil	Dilates the pupil

138. In pupillary light reflex:

Afferent: Optic nerve.

Efferent: Oculomotor nerve.

139. Questions (139), (140) and (140):

The optic disc contains no photoreceptors, it is the blind spot of the retina, and the macula lutea is located lateral to the optic disc. The Foveola (fovea centralis) is a located at the center of the macula lutea, it is the area of most acute vision (high visual resolution).

140. See answer (139) above.

141. See answer (139) above.

142. The main site that is responsible for refraction of the light is the Cornea.

143. The lens is adjusted for near vision by its increased convexity due to the contraction of the ciliary muscle.

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Section Two: Scenario-Based MCQs

Clinical Case Scenario (1)

144. Questions (144) and (145): The mandible articulates with the mandibular fossa of the temporal bone through its condylar process to form the temporomandibular joint (TMJ). It is a typical synovial joint, the most common type of its dislocation is anterior dislocation.

145. See answer (144) above.

146. In general, Most of the mandibular fractures occur as follow: in order from higher to lower frequencies: body (29%), condyloid process (26%), angle of the mandible (25%), symphyses (17%), ramus (4%) and coronoid process (1%).

Sites of mandibular fractures vary according to the cause:

Automobile accidents, the most common site of fracture is the condylar region.

During **fights**, the most common site of fracture is the angle of the mandible.

147. The most commonly injured nerve following mandibular fracture is the inferior alveolar nerve.

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Clinical Case Scenario (2)

148. Questions (148), (149) and (145):

The internal jugular vein is the best for central venous access, in comparison with the subclavian vein: the risk of development of pneumothorax or thrombotic complications is less, as well as the risk of arterial injury, its location is constant in the neck. The right internal jugular vein is favored than the left because it is larger than the right, has shorter and straighter course to the heart (right atrium).

Surface marking and important relations:

Superiorly: it starts as a continuation of the sigmoid sinus, at a point between the mastoid process and the angle of the mandible, runs downwards as part of the carotid sheath, the common carotid and the internal carotid arteries are located medial to the internal jugular vein.

Inferiorly: it ends at the level of T1, deep to the sternoclavicular joint. The vein is between the clavicular and sternal heads of the sternocleidomastoid muscle, its pulsation can be seen just superior to the medial third of the clavicle.

Catheterization:

Posterior approach: the needle is inserted above the clavicle at the posterior border of the sternocleidomastoid muscle.

Anterior approach: The needle is inserted at the apex of the triangle formed by the clavicular and sternal heads of the sternocleidomastoid muscle (lesser supraclavicular fossa), in a caudal direction.

149. See answer (148) above.

150. See answer (148) above.

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Clinical Case Scenario (3)

151. Questions (151), (152), (153) and (154): The following are the groups of lymph nodes of the head and neck, they all eventually drain into the deep cervical lymph nodes.

Lymph node	Location	Receive from
Occipital	Close to the occipital bone posteriorly	Posterior part of the SCALP
Mastoid	Close to the mastoid process, posterior to the ear	SCALP, auricle and external auditory meatus
Parotid	Close to the parotid gland	SCALP, parotid gland, auricle and external auditory meatus
Buccal	The cheek, over Buccinator	Face
Submandibular	Submandibular gland (superficially)	SCALP (anterior part), side of the face, part of the tongue and floor of the mouth.
Submental	Submental triangle	Tip of the tongue and anterior part of the oral cavity: (incisor teeth, middle of the lower lib and skin of the chin
Anterior cervical	Close to the anterior jugular vein	Anterior part of the neck (skin and superficial tissue)
Superficial cervical	Close to the external jugular vein	Area around the angle of the mandible, part of the parotid gland and ear
Retropharyngeal	Posterior to the pharynx	Mainly the nasal cavity
Laryngeal	Anterior to the larynx	Larynx
Paratracheal	Close to the trachea	Trachea and thyroid gland

The **deep cervical lymph nodes** are in the carotid triangle, they form a chain that is located close to the internal jugular vein within the fascia of the carotid sheath, deep to sternocleidomastoid muscle.

The lymphatic vessels from the deep cervical nodes form lymphatic trunks, the jugular lymphatic trunks, which drain into the junction of the internal jugular and subclavian veins on the right side and the thoracic duct on the left side.

Metastases from the thorax and abdomen can involve the deep cervical lymph nodes along the transverse cervical artery.

152. See answer (151) above.

153. See answer (151) above.

154. See answer (151) above.

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Section Three: Extended Matching Questions (R-Type items)

Skull foramina

155. Optic nerve (a).
156. Facial nerve (f).
157. Trochlear nerve (g).
158. Vertebral artery (d).
159. Ophthalmic vein (g).
160. Ophthalmic artery (a).
161. Mandibular nerve (b).
162. Ophthalmic nerve (g).
163. Oculomotor nerve (g).
164. Middle meningeal artery (e).
165. Glossopharyngeal nerve (c).
166. Accessory meningeal artery (b).

Triangles of the Neck

167. Facial vein (e).
168. Vagus nerve (a).
169. Accessory nerve (b).
170. Subclavian artery (f).
171. Thyrohyoid muscle (c).
172. Sternohyoid muscle (c).
173. Internal jugular vein (a).
174. Anterior jugular vein (d).

Motor Innervation of the Head and Neck

175. Mylohyoid (d).
176. Stylohyoid (a).
177. Geniohyoid (f).
178. Orbicularis Oris (a).
179. Occipitofrontalis (a).
180. Medial Pterygoid (d).
181. Tensor Veli Palatine (d).
182. Stylopharyngeus (h).
183. Vocalis (j).
184. Posterior cricoarytenoid (j).
185. Stapedius (a).

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186. Tensor tympani (d).
187. Lateral rectus (b).
188. Superior oblique (c).
189. Medial rectus (e).
190. Cricothyroid (i).

Connections to the nasal cavity

191. Frontal sinus (a).
192. Maxillary sinus (a).
193. Sphenoidal sinus (d).
194. Posterior ethmoidal sinus (c).
195. Middle ethmoidal air cells (a).
196. Anterior ethmoidal air cells (a).

Functions of the Ear

197. Sound waves collection (a)
198. Inhibition of excessive movement of Stapes (b).
199. Reduction of the amplitude of Malleus oscillations (c).
200. Equalization of the pressure on both sides of the eardrum (d)

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Section Four: Medical Imaging

Figure (1): Lateral Cervical Radiograph:

This is a lateral cervical radiograph, identify the following structures:

201. This soft tissue structure (A): epiglottis.

202. This bony structure (B): body of the 3rd cervical vertebra.

Figure (2): Anteroposterior Cervical Radiograph:

This is anteroposterior cervical radiograph, identify the following structures:

203. This bony projection (A): Spinous process of the 1st thoracic vertebra

204. This soft structure (B) "hint : filled with air": trachea

Figure (3): MRI Cervical Spine:

This is MRI cervical spine, identify the following structures:

205. This structure (A): Oropharynx

206. This bony structure (B): Odontoid process.

207. This structure (C): Intervertebral disc space (C6/C7)

Figure (4): CT Brain - Bone Window:

This is CT brain - bone window, identify the following structures:

208. This bony structure (A): Right zygomatic bone.

209. This bony structure (B): Right ethmoidal air cells.

210. This bony structure (C): Left mastoid air cells.

Figure (5): CT PNS (Paranasal Sinuses), Coronal view

This is a CT PNS (Paranasal Sinuses), Coronal view, identify the following structures:

211. This air-containing cavity (A): right maxillary sinus

212. This bony structure (B): left inferior nasal turbinate (concha).

Figure (6): 3D reconstruction of CT-Skull: Skull and facial bones

This is a 3D Reconstruction: Skull and facial bones, identify the following sutures: structures:

213. (A): Right coronal suture.

214. (B): Left frontozygomatic suture.

Figure (7): 3D reconstruction of CT-Skull:

This is a 3D Reconstruction: Skull: posterior view, identify the following: structures:

215. Suture (A): Left lambdoid suture

216. Bony projection (B): External occipital protuberance.

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Figure (8): 3D reconstruction of CT-Skull:

This is a 3D Reconstruction: Skull: lateral view, identify the following sutures: structures:

217. Bony structure (A): Left coronoid process.

218. Bony structure (B): Left mastoid process.

219. This bone (C): Left hyoid bone.

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